



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
GRADE SEVEN EXAMINATION

MATHEMATICS
PAPER 1

002/1

OCTOBER 2017 SESSION

Time: 2 hours

INSTRUCTIONS TO CANDIDATES

1. Read **all** the instructions carefully.
2. **Do not** open this booklet until you are told to do so by the invigilator.
3. Use only an HB pencil for all entries on the answer sheet.
4. When you are told to start, choose **one** correct answer from the suggested answers and shade it **very dark** as shown in the examples at the top of the answer sheet.
5. If you wish to change your answer, **erase** it **completely** with a pencil rubber and then shade the new choice.
6. If **more** than **one** box is shaded for any one answer, that answer will be regarded as **wrong**.
7. **If you do not understand** the instructions, **ask** the invigilator to explain them to you **before you start**.
8. Answer **all** the questions on the separate answer sheet provided.
9. Paper for rough work will be provided.

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1

Thirty five thousand six hundred and one in figures is

- A 350 601.
- B 305 601.
- C 35 601.
- D 35 006.

2

16,325 correct to one decimal place is

- A 1632, 5.
- B 163,3.
- C 16,3.
- D 1,6.

3

$20 - 8 \times 2$

- A 10.
- B 8.
- C 5.
- D 4.

4

$\frac{18}{4}$ expressed as a mixed number is

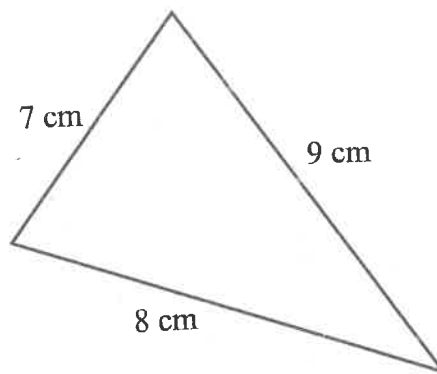
- A $1\frac{1}{2}$.
- B $2\frac{1}{2}$.
- C $3\frac{1}{4}$.
- D $4\frac{1}{2}$.

5

How many lines of symmetry does a regular octagon have?

- A 8.
- B 6.
- C 4.
- D 2.

- 6 The special name of the triangle below is



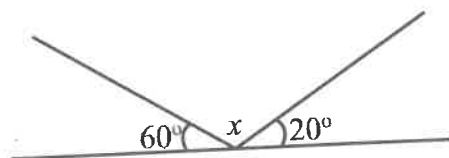
- A an isosceles.
B a scalene.
C an equilateral.
D a right angled.
- 7 $\frac{7}{8}$ as a decimal is
- A 0,975.
B 0,875.
C 0,78.
D 0,675.
- 8 If $24 \div 8 = 3$ then $0,24 \div 0,8 =$
- A 30.
B 3.
C 0,3.
D 0,03.
- 9 What is the product of 7,4 and 0,02?
- A 148.
B 14,8.
C 1,48.
D 0,148.

- 10 What is the missing number in the sequence 2; 5; 9; -----; 20?
- A 12.
B 16.
C 14.
D 19.
- 11 What is the difference between $3 \times (8 + 4)$ and $3 \times 8 + 4$?
- A 8.
B 15.
C 42.
D 75.
- 12 $\frac{2}{5}$ of a number is 30. The number is
- A 12.
B 15.
C 42.
D 75.
- 13 $0,018 \div 0,6 =$
- A 0,003.
B 0,03.
C 0,3.
D 3,0.
- 14 What is the sum of 8; 4,326 and 0,2?
- A 12,526.
B 12,428.
C 12,328.
D 4,336.

15 $\frac{10}{15} = \frac{n}{3}$. Therefore n is

- A 4.
- B 3.
- C 2.
- D 1.

16 The angle marked x on the diagram below is



- A 40° .
- B 60° .
- C 80° .
- D 100° .

17 $\left(3\frac{1}{2} - 2\frac{2}{5}\right) \times 7\frac{1}{2} =$

- A $1\frac{1}{5}$.
- B 7.
- C $8\frac{1}{4}$.
- D $8\frac{1}{15}$.

18 A baby's milk bottle holds 375 cm^3 . How many times can it be filled from a 3 litre flask?

- A 9.
- B 8.
- C 90.
- D 80.

- 19 The table shows the measurements of four boxes. P, Q, R and S

	Length	Width	Height
P	7 cm	5 cm	2 cm
Q	6 cm	3 cm	5 cm
R	5 cm	2,5 cm	4 cm
S	5 cm	4 cm	2 cm

The box with the greatest volume is

- A** Q.
- B** P.
- C** R.
- D** S.

- 20 The sum of 8^2 and 9^2 is

- A** 17.
- B** 34.
- C** 97.
- D** 145.

- 21 Twenty five minutes past seven in the evening in 24 hour notation is

- A** 0725.
- B** 1725.
- C** 1925.
- D** 1935.

- 22 If $m \times 12 = 132$, then m is

- A** 11.
- B** 12.
- C** 22.
- D** 33.

23 The Lowest Common Multiple (LCM) of 6 and 9 is

- A 6.
- B 12.
- C 18.
- D 54.

24 8,099 to the nearest whole number is

- A 8.
- B 9.
- C 81.
- D 8099.

25 A suit costs \$60. If 10 % deposit is paid, then the balance is

- A \$6.
- B \$50.
- C \$54.
- D \$66.

26 There are 350 pupils in a school. If the ratio of boys to girls is 4 : 3, then the number of boys in the school is

- A 50.
- B 100.
- C 150.
- D 200.

27 A motorist covered a distance of 90 km in 45 minutes. What was the average speed?

- A 120 km/hr.
- B 100 km/hr.
- C 90 km/hr.
- D 45 km/hr.

28 The number of 250 g packets of washing powder that can be packed from a 25 kg bag are

- A 10.
- B 50.
- C 100.
- D 1 000.

- 29** 10 marks out of 20 as a percentage is
- A** 50 %.
 - B** 30 %.
 - C** 20 %.
 - D** 10 %.
- 30** Lessons start at 8.00 a.m. A girl arrived at school 13 minutes late. She arrived at
- A** 7.13 a.m.
 - B** 7.47 a.m.
 - C** 8.13 a.m.
 - D** 8.47 a.m.
- 31** In 1960 a boy was 7 years old. In 2006 his age was
- A** 46.
 - B** 53.
 - C** 63.
 - D** 66.
- 32** The cost of painting a wall measuring $4\frac{1}{2}$ m by 8 m at \$3,15 per square metre is
- A** \$11,34.
 - B** \$11,30.
 - C** \$113,40.
 - D** \$1134,00.
- 33** Twelve men can build a kraal in 30 hours. How long can it take 8 men to build the kraal working at the same rate?
- A** 45 hours.
 - B** 20 hours.
 - C** 15 hours.
 - D** 10 hours.
- 34** In 2008 the total number of days for February, March and April was
- A** 88.
 - B** 89.
 - C** 90.
 - D** 91.

35. The table below shows minimum temperatures recorded at a school.

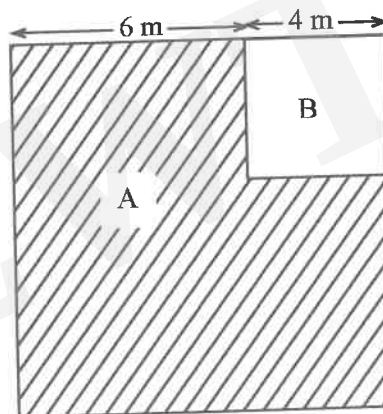
Day	Sun	Mon	Tue	Wed	Thur	Fri	Sat
Minimum Temperature °C		17	18	15	12	19	18

The average minimum temperature for the whole week was 16 °C. The temperature for Sunday was

- A 13 °C
B 14 °C
C 14,5 °C
D 15 °C
36. $10 - 16 + 8 =$

- A 14.
B 8.
C 4.
D 2.

37. In the diagram below, B is a square in square A.

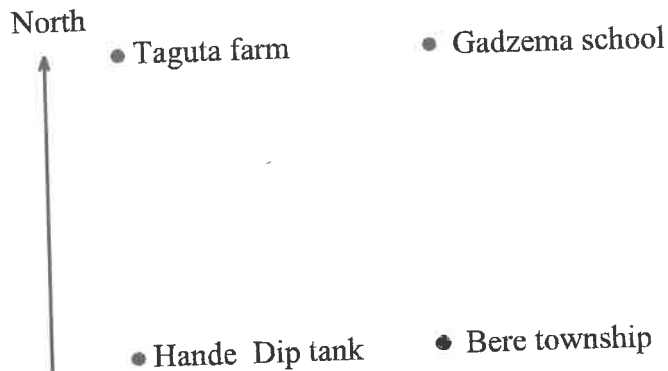


The area of the shaded part is

- A 16 m².
B 36 m².
C 84 m².
D 100 m².
38. What is the difference between the product of 40 and 10 and the sum of 40 and 10?
- A 450.
B 360.
C 350.
D 340.

- 39** The interest on \$ 600 invested for 2 years at 20 % per annum is
- A** \$ 1 200.
 - B** \$ 240.
 - C** \$ 120.
 - D** \$ 40.
- 40** $1\frac{2}{5} \times 2\frac{3}{7} =$
- A** $3\frac{1}{7}$.
 - B** $3\frac{1}{5}$.
 - C** $3\frac{2}{5}$.
 - D** $3\frac{17}{35}$.
- 41** A one hectare piece of land was divided into 8 equal plots. [1 hectare = 10 000 m²]. The area of each plot was
- A** 1 000 m².
 - B** 1 250 m².
 - C** 1 500 m².
 - D** 1 750 m².
- 42** A train left Harare at 2030 and arrived in Bulawayo at 0605 the next day. How long did the journey take?
- A** 3 hours 30 minutes
 - B** 6 hours 5 minutes.
 - C** 7 hours 40 minutes.
 - D** 9 hours 35 minutes.

43



In the diagram above, the direction of the township from the school is

- A North.
- B East.
- C West.
- D South.

44

The gross mass of a tin of beans is 2 085 g. The empty tin weighs 250 g. The mass of beans is

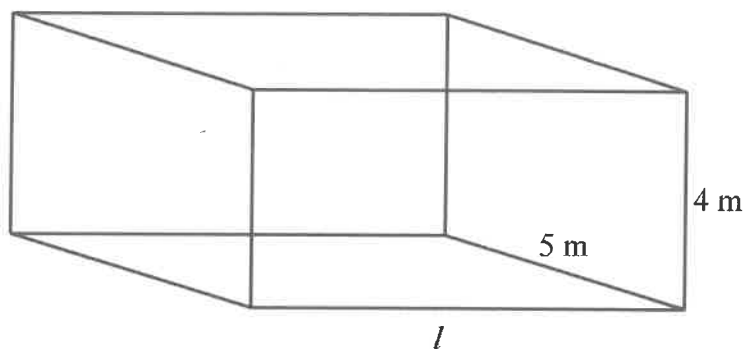
- A 250 g.
- B 1 835 g.
- C 2 235 g.
- D 2 335 g.

45

A farmer weeded $\frac{2}{5}$ of his field on Monday and $\frac{1}{3}$ on Tuesday. What fraction of the field remained unweeded?

- A $\frac{11}{15}$
- B $\frac{4}{15}$
- C $\frac{1}{5}$
- D $\frac{1}{15}$

- 46 A rectangular tank has a volume of 120 m^3 . It is 5 m wide and 4 m deep.



The length, l , of the tank is

- A 6 m.
B 20 m.
C 24 m.
D 30 m.
- 47 A family has four sons, John, Jimmy, Jairs and Jethro, whose dates of birth are shown in the table below.

Name	John	Jimmy	Jairus	Jethro
Date of birth	16.02.81	04.04.96	21.10.72	14.09.91

Who is the youngest of the four sons?

- A John.
B Jethro.
C Jairus.
D Jimmy.
- 48 How many decades are in a century?
- A 10.
B 100.
C 1 000.
D 10 000.

The chart below shows the distances between selected towns in Zimbabwe. Study the chart and use it to answer question 49 and question 50.

Beitbridge			
322	Bulawayo		
607	279	Kadoma	
472	164	135	Gweru

49 The distance from Beitbridge to Kadoma is

- A 607.
- B 472.
- C 279.
- D 135.

50 A bus took 2 hours to travel from Bulawayo to Gweru. What was its average speed?

- A 164 km/h.
- B 135 km/h.
- C 67,5 km/h.
- D 82 km/h.



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
GRADE SEVEN EXAMINATION

MATHEMATICS
PAPER 2

002/2

OCTOBER 2017 SESSION

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INSTRUCTIONS TO CANDIDATES

1. Answer **all** questions in Section **A** on the separate answer paper provided.
2. Answer any **three** questions in Section **B** on the separate answer paper provided.
3. Section **A** carries 25 marks.
4. Section **B** carries 15 marks.
5. To obtain full marks for any question, **all working** must be shown.
6. Do **not** measure from given diagrams.
7. Electronic calculators and slide rules **must not** be used in the examination.

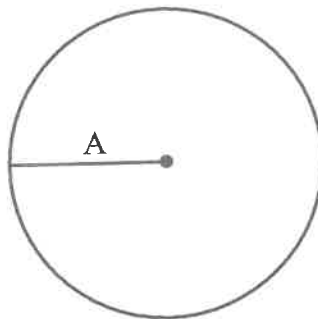
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Section A (25 marks)

Answer all questions in this section.

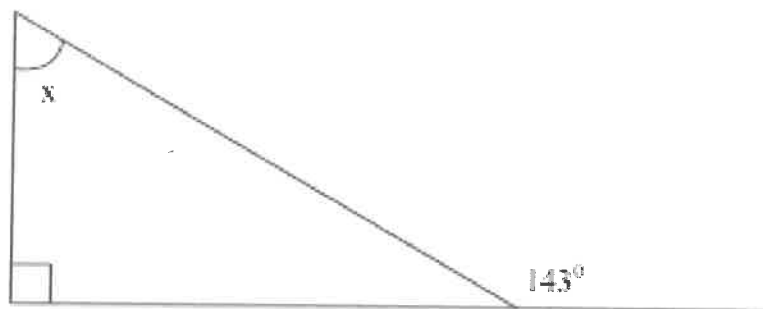
- 1 (a) Write two thousand and twenty in figures. [1]
(b) Round off 10,78 to the nearest whole number. [1]
- 2 (a) Find the sum of 123; 74 and 0, 3. [2]
(b) Express 250 g as a fraction of 1 kg, in its lowest terms. [2]
- 3 Simplify:
(a) $2\frac{1}{5} + 2\frac{1}{3}$ [2]
(b) $3\frac{3}{4} \times 2\frac{2}{3}$ [2]
- 4 Mother brought home 2 640 sweets. She shared 1 985 sweets equally among her 5 children.
(a) How many sweets did each child get? [2]
(b) Calculate the number of sweets not shared. [2]
- 5 1245 is the time in the 24 hour notation.
Express the time in 12 hour notation. [1]
- 6 (a) The diagram below shows a circle with the centre marked by a dot.



Name line A in the circle.

[1]

- (b) Below is a right-angled triangle, with angles as shown.



Find the angle marked x in the diagram. [3]

- (c) Calculate the area of a triangle with base 8 cm and height 5 cm. [2]

- 7 A woman bought 17 soap tablets each weighing 200 g. They were put in a box with a net mass of 750 grams.

(a) Calculate the mass of the 17 soap tablets. [2]

(b) Find the gross mass of the box. [2]

Section B (15 marks)

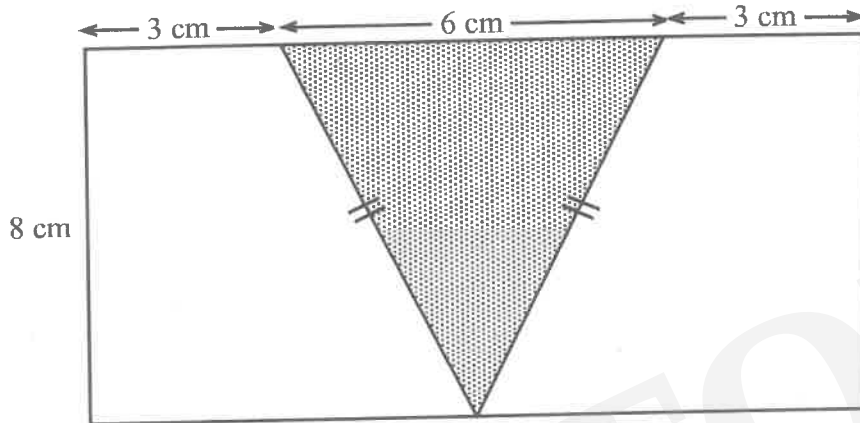
Answer any *three* questions from this section.

8

Mr Sibanda fenced his garden using 5 strand barbed wire. The garden is 30 m long and 15 m wide.

- (a) Find the perimeter of the garden. [3]
- (b) Calculate the length of the barbed wire needed to fence the garden. [2]

9



The diagram above shows a rectangle. Part of the rectangle is a shaded triangle.

- (a) Calculate the area of the shaded part. [2]
- (b) Calculate the area of the unshaded part. [3]

10

The table below shows vegetables sold from a school garden.

vegetable	Crates	unit price	total cost
tomatoes	4	\$1,80	\$7,20
cucumbers		\$4,20	\$16,80

- (a) How many crates of cucumbers were sold? [3]
- (b) Calculate the total cost of 8 crates of tomatoes. [2]

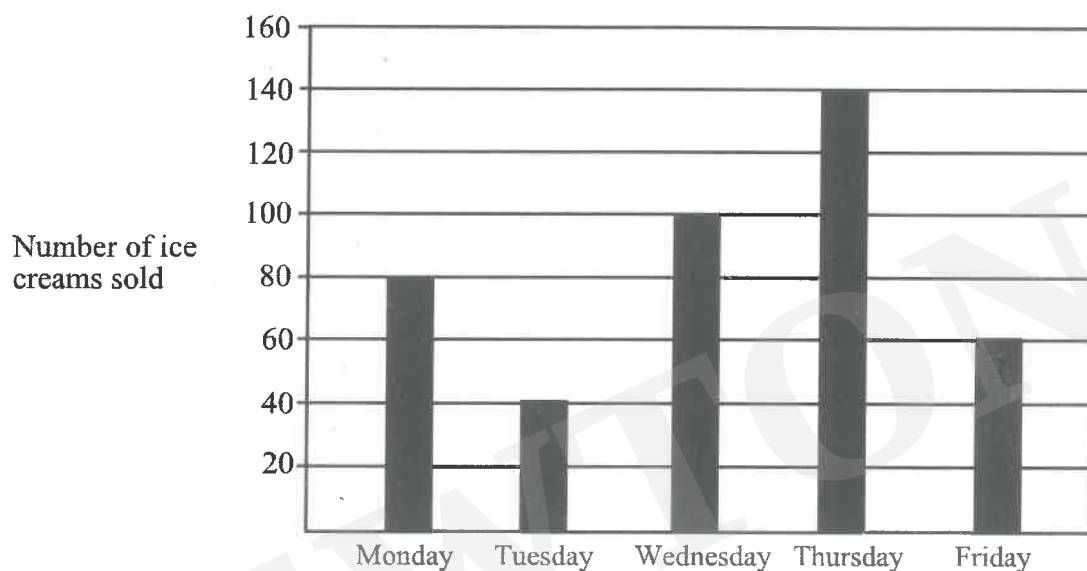
- 11 In a big box, there are 12 cartons of matches. A carton has 10 small boxes of matches.

(a) How many small boxes of matches are in the big box? [2]

(b) Tom, Mary and Thabiso shared the small boxes in the ratio 2 : 3 : 5, respectively.

Calculate Mary's share. [3]

- 12 The graph below shows the number of ice creams sold by a vendor from Monday to Friday.



(a) How many ice creams did the vendor sell in the 5 days? [2]

(b) The ice creams were sold at 35 c each.

How much did the vendor get in the 5 days? Give your answer in dollars. [3]



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MATHEMATICS
PAPER 1

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1 515 042 rounded off to the nearest hundred is

- A 500 000.
- B 515 000.
- C 515 040.
- D 515 100.

2 The value of the underlined digit in 73 256 is

- A 50.
- B 500.
- C 5 000.
- D 50 000.

3 6 count on 4 is

- A 2.
- B 10.
- C 24.
- D 64.

4 $\frac{3}{4}$ of 52 =

- A 3
- B 13
- C 16
- D 39

5 12.36 a.m. in 24 hour notation is

- A 1236.
- B 1224.
- C 0036.
- D 2436.

6 $0,7 \times 3,2 =$

- A 22,4.
- B 21,40.
- C 2,24.
- D 2,14.

7 The best unit for measuring the length of a classroom is the

- A mm.
- B cm.
- C m.
- D km.

8 What is the size of one revolution?

- A 90°
- B 180°
- C 270°
- D 360°

9 If the multiplicand is 7 and the product is 63, the multiplier is

- A 441.
- B 70.
- C 56.
- D 9.

10 A class has 17 girls and 23 boys. Half the class has _____ pupils.

- A 6
- B 15
- C 20
- D 40

11 $\frac{123}{30}$ expressed as a mixed number is

- A $3\frac{3}{10}$
- B $4\frac{1}{30}$
- C $3\frac{1}{10}$
- D $4\frac{1}{10}$

12 $14 \times 12 =$

- A 26
- B 42
- C 128
- D 168

13 How many sides does a decagon have?

- A 5
- B 8
- C 7
- D 10

- 14 Which of the following is a leap year?
- A 1994
 - B 1998
 - C 2002
 - D 2004
- 15 $8 + 8^2 \div 4 =$
- A 18
 - B 24
 - C 64
 - D 72
- 16 John, Chipu and Ruth shared \$160 in the ratio 3:5:2, respectively. What was Chipu's share?
- A \$80
 - B \$48
 - C \$32
 - D \$16
- 17 What is the perimeter of a square with an area of 49 m^2 ?
- A 42
 - B 28
 - C 14
 - D 7
- 18 437 increased by 46 is
- A 897
 - B 483
 - C 473
 - D 391
- 19 $4 - 6 + 8 =$
- A 2
 - B 6
 - C 10
 - D 18
- 20 $1\frac{1}{2} + 3\frac{3}{4} =$
- A $5\frac{1}{4}$
 - B $4\frac{1}{4}$
 - C $5\frac{2}{3}$
 - D $4\frac{2}{3}$

- 21 3 men took 4 days to complete a job. How long will it take 4 men to do the same job working at the same rate?

A 16 days
B 12 days
C 7 days
D 3 days

22 $0,2 \times 0,2 \times 0,2 =$

A 8.
B 0,8.
C 0,08.
D 0,008.

- 23 Arrange the fractions below in descending order $\frac{1}{2}, \frac{2}{5}, \frac{1}{10}, \frac{1}{4}$.

A $\frac{1}{4}, \frac{1}{10}, \frac{2}{5}, \frac{1}{2}$

B $\frac{2}{5}, \frac{1}{2}, \frac{1}{10}, \frac{1}{4}$

C $\frac{1}{10}, \frac{1}{4}, \frac{2}{5}, \frac{1}{2}$

D $\frac{1}{2}, \frac{2}{5}, \frac{1}{4}, \frac{1}{10}$

- 24 How many lines of symmetry does an equilateral triangle have?

A 1
B 2
C 3
D 4

- 25 $\frac{75}{100}$ as a fraction in its lowest terms is

A $\frac{1}{2}$.
B $\frac{7}{10}$.
C $\frac{3}{4}$.
D $\frac{15}{20}$.

26 35 in Roman numerals is

- A XLV.
- B XXLV.
- C XXXV.
- D XXXIII.

27 The volume of the box is 80 cm^3 . It has a length of 5 cm and a height of 4 cm.

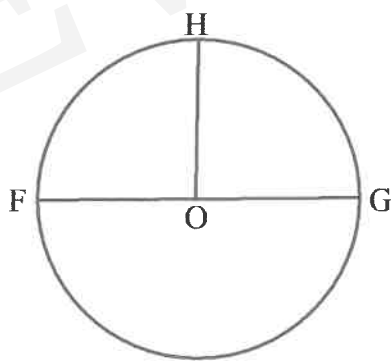
What is the width of the box?

- A 4
- B 16
- C 20
- D 1600

28 $\frac{4}{100}$ as a decimal number is

- A 4,0.
- B 0,4.
- C 0,04.
- D 0,004.

29 Below is a circle, with centre at O.



If line $HO = 6,5 \text{ cm}$, then line $FG =$

- A 65 cm
- B 13 cm
- C 12 cm
- D 7 cm

- 30 Mr Khami spends 35 % of his salary on rent, 25% on school fees and 15 % on transport.

What percentage of his salary is left?

- A 25 %
- B 35 %
- C 75 %
- D 100 %

31 $0,4 + 1 + 2,25 =$

- A 2,30
- B 3,29
- C 2,66
- D 3,65

32 $1\ 001 - 899 =$

- A 212
- B 112
- C 102
- D 11

- 33 Tadiwa was facing North East and he turned anticlockwise through 270° . He is now facing

- A South East.
- B South West.
- C North East.
- D North West.

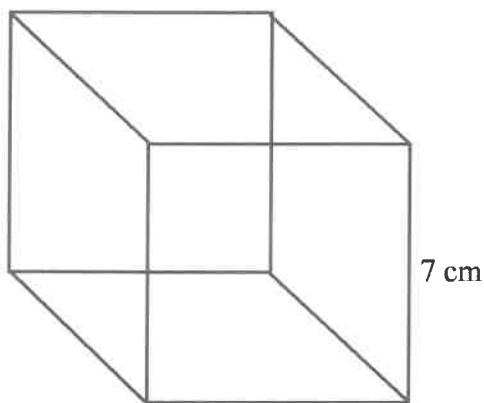
34 $\frac{3}{4} \times \frac{1}{3} =$

- A $\frac{1}{4}$
- B $\frac{1}{12}$
- C $\frac{4}{9}$
- D $\frac{3}{4}$

- 35 The HCF of 30 and 45 is

- A 90.
- B 75.
- C 15.
- D 5.

- 36 A loaded truck has a gross mass of 4 tonnes. If the truck weighs 1,4 tonnes, the net mass of the load is
- A 5,4 tonnes.
B 2,6 tonnes.
C 1,4 tonnes.
D 4 tonnes.
- 37 30 % of \$350 is
- A \$380.
B \$320.
C \$245.
D \$105.
- 38 Timmy left home at 9.30 a.m. and arrived in Masvingo at 2.30 pm, on the same day. His journey took
- A 7 hours.
B $6\frac{1}{2}$ hours.
C 6 hours.
D 5 hours.
- 39 A man bought a stove marked \$950 and was given 20 % discount. The amount he paid for the stove was
- A \$1 140
B \$930
C \$760
D \$190
- 40 A cube has a height of 7 cm.



The volume of the cube is

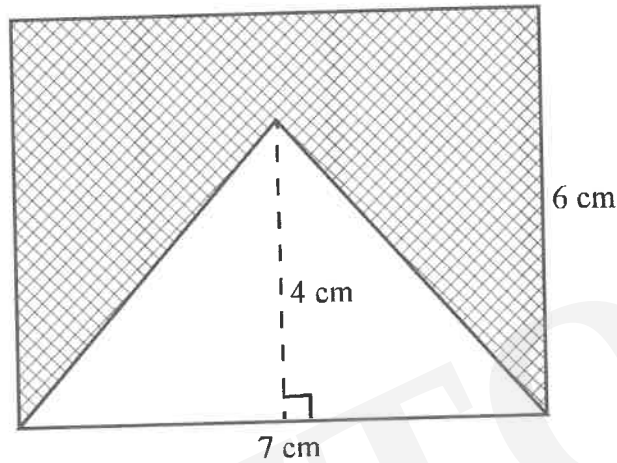
- A 21 cm^3 .
B 49 cm^3 .
C 56 cm^3 .
D 343 cm^3 .

- 41 Jabulani's bank account had \$150 and he deposited \$75. After 2 days he withdrew \$85. How much was left in the account?

A \$225
B \$140
C \$310
D \$160

Use the diagram below to answer question 42 and question 43.

The diagram shows a triangle inside a rectangle.



- 42 The area of the triangle is

A 28 cm^2
B 21 cm^2
C 14 cm^2
D 11 cm^2

- 43 The area of the shaded part is

A 42 cm^2
B 28 cm^2
C 21 cm^2
D 13 cm^2

- 44 $\frac{1}{6} + \frac{3}{5} - \frac{2}{3}$ is

A $\frac{1}{4}$
B $\frac{1}{10}$
C $\frac{11}{30}$
D $\frac{23}{30}$

- 45 Marks of six grade seven pupils were 9; 6; 5; 10, 15 and 3.

What was their average mark?

- A 6
- B 8
- C 16
- D 48

46 $0,625 \div 0,5 =$

- A 1,25
- B 12,5
- C 125
- D 0,125

- 47 A lorry travelled 130 km in two hours.

At what speed was it travelling?

- A 65 km/h
- B 128 km/h
- C 132 km/h
- D 260 km/h

48 $8^3 =$

- A 512
- B 83
- C 24
- D 11

The table below shows the number of spellings given to Grade 5 pupils on each day of the week. Use the table to answer question 49 and question 50.

Day	Number of spellings
Monday	20
Tuesday	25
Wednesday	15
Thursday	25
Friday	30

- 49 How many more spellings were written on Friday than on Wednesday?

- A 25
- B 45
- C 30
- D 15

50 Which day has the same number of spellings as Thursday?

- A** Friday
- B** Monday
- C** Tuesday
- D** Wednesday

NEWTON



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GRADE SEVEN EXAMINATION, 2018.

MATHEMATICS
PAPER 2

002/2

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3. Section **A** carries 25 marks.
4. Section **B** carries 15 marks.
5. Omission of essential working may result in loss of marks.
6. Do **not** measure from given diagrams.
7. Electronic calculators and slide rules **must not** be used in the examination.

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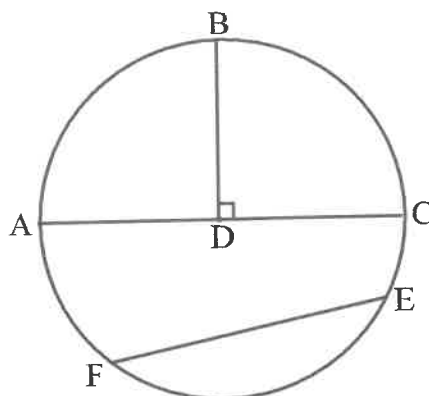
Section A (25 marks)

*Answer **all** questions in this section.*

1. (a) Write three thousand and four in figures. [1]
- (b) Write down the number shown on the abacus below. [1]

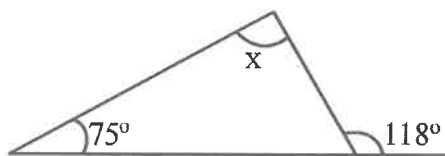
TTH	TH	H	T	U
XX	XXX	XX	XX	XX
XX	XXX		XX	X
X				
XX	XX	XXX	XX	XXX
XX	X	XXX	X	XXX
		X	XX	

2. (a) Express 0,35 as a common fraction in its lowest terms. [2]
- (b) Find the value of $6^4 + 4^0$. [2]
3. Express
- (a) 7.46 p.m. in 24 hour notation. [1]
- (b) $\frac{3}{5}$ of a century in years. [2]
4. Below is a circle. Point D is the centre of the circle.



- (a) State the special name given to line EF. [1]
- (b) If arc CB is 4 cm, calculate the distance round the circle. [2]

5. Simplify $9 - 12 \div 6$ [2]
6. Simplify:
- (a) $4 - 1,37$ [2]
- (b) $4,3 + 0,04$ [2]
7. The full marks for a spelling test was 20 marks. Peter got 75% in the test. Calculate Peter's mark out of 20. [2]
8. Mr Dube bought a suit marked \$200. He got 5% discount. Calculate the amount he paid. [2]
9. Below is a diagram with angles as shown.



Find the value of x .

[3]

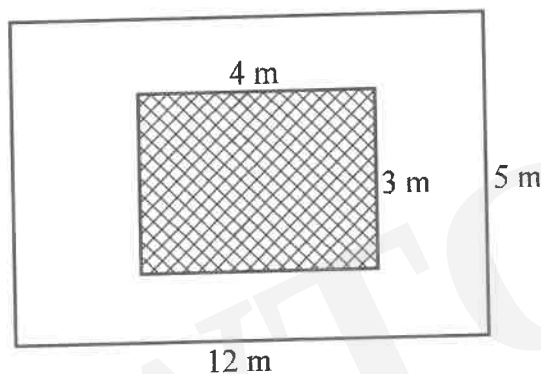
Section B (15 marks)

*There are five questions in this section. Answer any **three**.*

- 10.** There were 45 children in a class. $\frac{8}{9}$ of them entered a painting competition. One quarter of those who entered the competition won prizes.

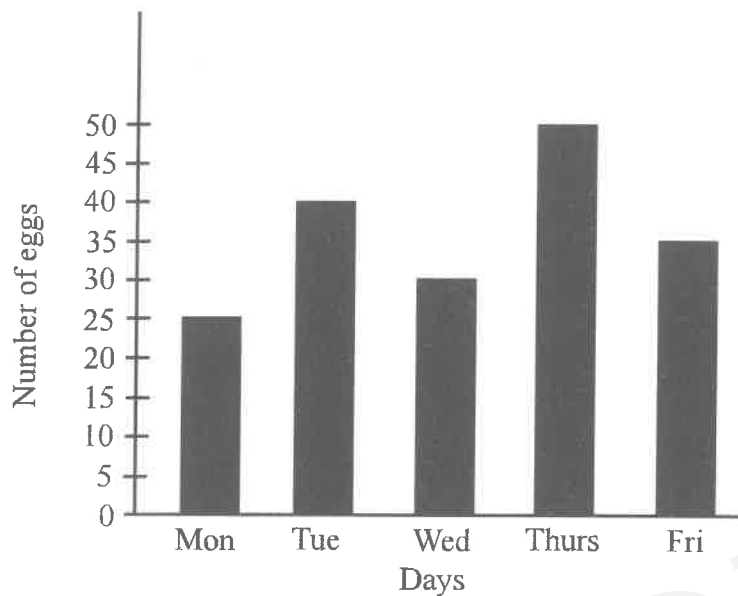
- (a) Find the fraction of children who did not enter the competition. [2]
- (b) Calculate the number of children who won prizes. [3]

- 11.** Below is a shape made up of a rectangle inside a bigger rectangle.



- (a) Find the difference between the length and width of the outer rectangle. [2]
- (b) Calculate the area of the unshaded part of the shape. [3]
- 12.** Five men can take 9 days to complete a piece of job.
- (a) Find how long it will take 15 men, working at the same rate to do the same job. [2]
- (b) If these 15 men were paid \$20 each per day, how much will they be paid for the job altogether? [3]

13. The graph below shows the number of eggs laid by 5 hens in a week.



- (a) Which day had the most number of eggs laid? [1]
- (b) How many more eggs were laid on Friday than on Monday? [2]
- (c) Find the total number of eggs laid in the 5 days. [2]

- 14 Use the distance table below to answer the questions that follow. All distances are in kilometres.

Beitbridge			
322	Bulawayo		
582	438	Harare	
586	579	265	Mutare
759	437	876	1016
Victoria Falls			

- (a) Which is the furthest place from Mutare? [1]
- (b) Find the difference between the distance from Harare to Victoria Falls and the distance from Beitbridge to Victoria Falls. [2]
- (c) A bus took 6 hours to travel from Bulawayo to Harare. Calculate the average speed of the bus. [2]



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
GRADE SEVEN EXAMINATION, 2019.

MATHEMATICS
PAPER 1

002/1

OCTOBER 2019 SESSION

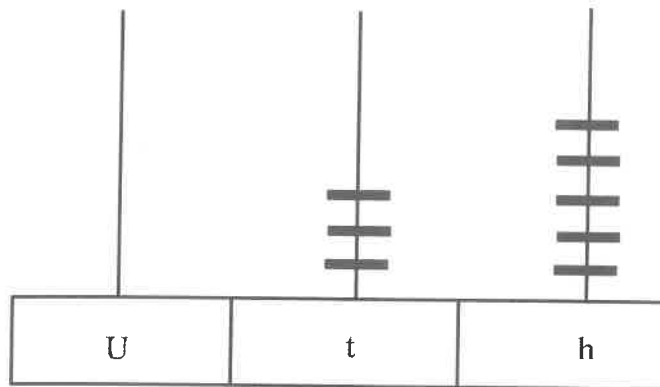
Time: 2 hours

INSTRUCTIONS TO CANDIDATES

1. Read **all** the instructions carefully.
2. **Do not** open this booklet until you are told to do so by the invigilator.
3. Use only an HB pencil for all entries on the answer sheet.
4. When you are told to start, choose **one** correct answer from the suggested answers and shade it **very dark** as shown in the examples at the top of the answer sheet.
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6. If **more than one** box is shaded for any one answer, that answer will be regarded as **wrong**.
7. **If you do not understand** the instructions, **ask** the invigilator to explain them to you **before you start**.
8. Answer **all** the questions on the separate answer sheet provided.
9. Paper for rough work will be provided.

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1



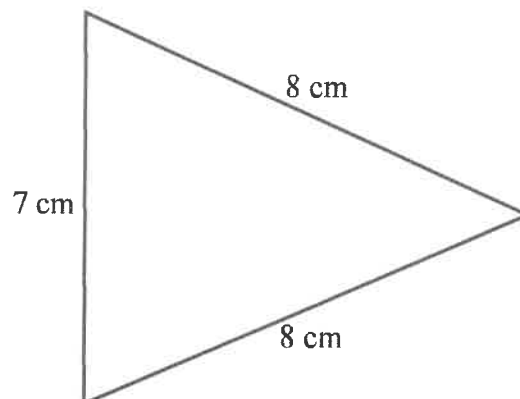
The number shown on the abacus above is

- A 3,50.
- B 5,30.
- C 0,35.
- D 0,53.

2 Which fraction is equivalent to $\frac{2}{3}$?

- A $\frac{8}{12}$
- B $\frac{4}{9}$
- C $\frac{1}{3}$
- D $\frac{3}{2}$

3 The type of triangle shown below is

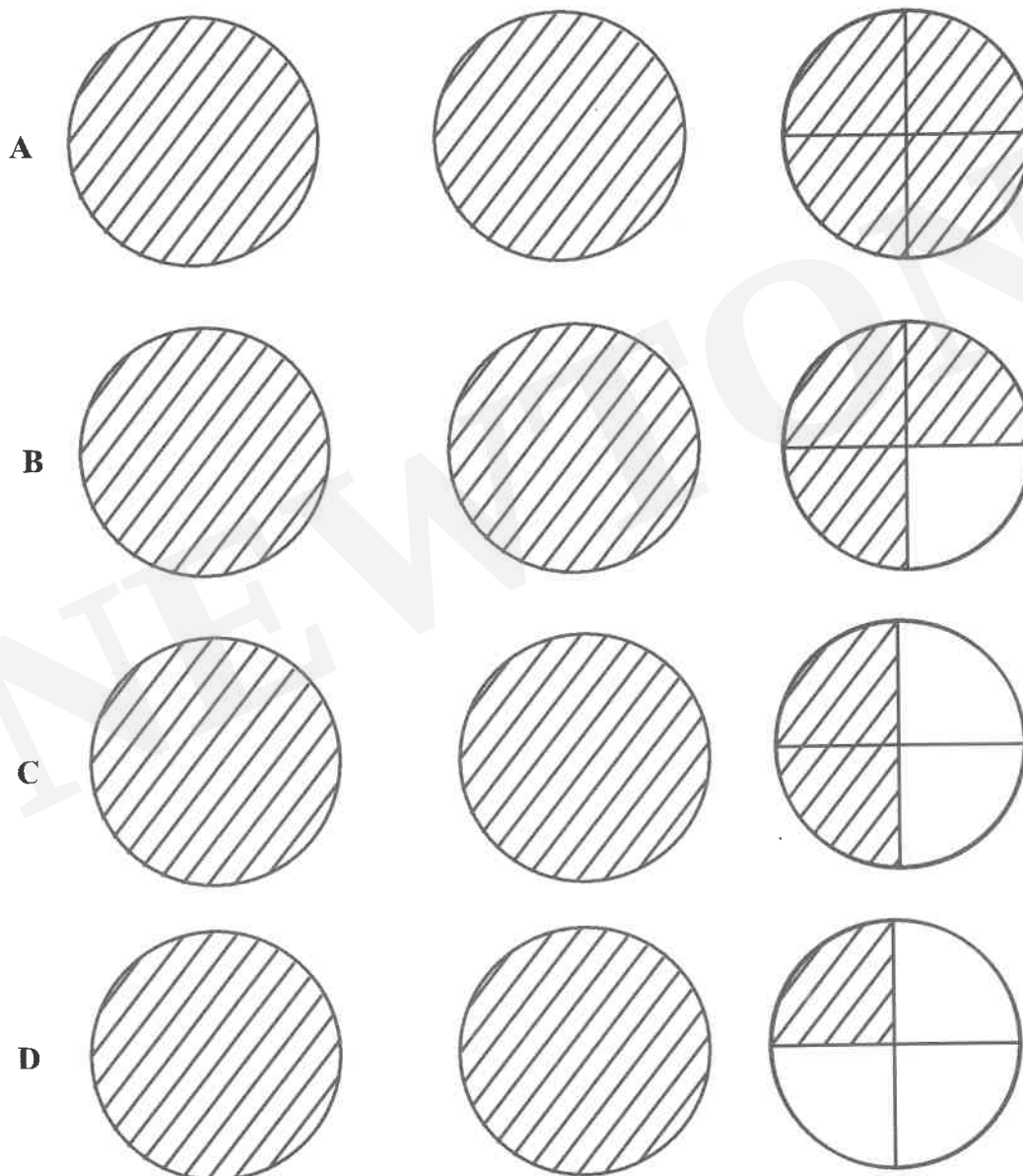


- A right angled.
- B equilateral.
- C isosceles.
- D scalene.

- 4 The total value of 2 fifty dollar notes, 10 fifty cent coins, 3 ten cent coins and 11 one cent coins is

A \$180,11.
 B \$150,41.
 C \$108,11.
 D \$105,41.

- 5 Which of the following shaded diagrams show $2\frac{3}{4}$?



6 $\frac{37}{7}$ as a mixed number is

A $3\frac{2}{7}$

B $5\frac{2}{7}$

C $5\frac{3}{7}$

D $7\frac{3}{7}$

7 A clock is 15 minutes ahead and it reads 8.12 a.m.
What is the correct time?

A 7.57 am

B 7.45 am

C 8.12 am

D 8.27 am

8 Express 80% as a fraction in its lowest terms.

A $\frac{80}{100}$

B $\frac{8}{10}$

C $\frac{4}{5}$

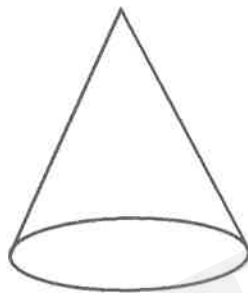
D $\frac{3}{5}$

9 Which of the following solid shapes is a cylinder?

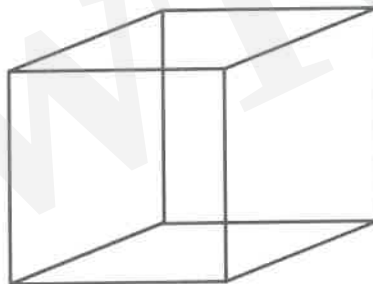
A



B



C



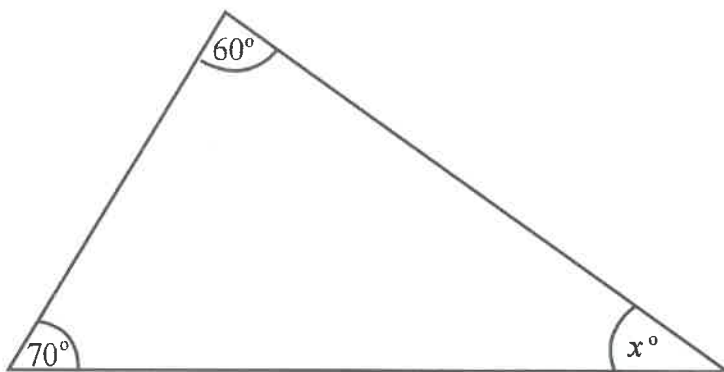
D



10 In $9 \times 6 = 54$ which one is a multiplicand?

- A** 54
- B** 15
- C** 9
- D** 6

- 11 What is the value of angle marked x in the diagram below?



- A 230°
 - B 130°
 - C 70°
 - D 50°
- 12 The distance right round a circle is called the

- A chord.
- B radius.
- C diameter.
- D circumference.

- 13 $697 \times 47 =$

- A 32 759.
- B 31 903.
- C 27 880.
- D 7 667.

- 14 Capacity is measured in

- A litres.
- B metres.
- C kilometres.
- D kilogrammes.

- 15 125 take away 69 is

- A 55.
- B 56.
- C 94.
- D 194.

- 16 The following is an incomplete calendar.

June 2015						
S	M	T	W	T	F	S
	1	2	3	4	5	6

On which day does the 15th of June 2015 fall?

- A Saturday
- B Monday
- C Thursday
- D Tuesday

17 $2 - 0,835 =$

- A 2,165.
- B 1,165.
- C 0,833.
- D 0,635.

18 $2,5 \times 100 =$

- A 0,25.
- B 2,50.
- C 250.
- D 2 500.

19 $\frac{3}{4} + \frac{1}{6} \div \frac{2}{3} - \frac{1}{8} =$

- A $\frac{1}{8}$.
- B $\frac{7}{8}$.
- C $1\frac{1}{4}$.
- D $1\frac{5}{8}$.

- 20 A cyclist covers 15 km in 45 minutes.
Find the average speed in km/h.

A $\frac{1}{3}$ km/h

B $\frac{3}{4}$ km/h

C 3 km/h

D 20 km/h

- 21 A class has 50 pupils, 20 of them are girls.
What is the percentage of boys in the class?

A 60%

B 50%

C 40%

D 20%

- 22 Hours Minutes

 9 20
- 3 40

A 5 hours 20 minutes

B 5 hours 40 minutes

C 6 hours 20 minutes

D 6 hours 40 minutes

- 23 A container holds 180 cm³ of water when $\frac{1}{5}$ full.
What is its capacity when full?

A 180 cm³

B 360 cm³

C 720 cm³

D 900 cm³

- 24 Mr Moyo bought a television set on hire purchase. He paid a deposit of \$40 and monthly instalments of \$16 for 24 months.
How much did he pay in total for the television set?

A \$56

B \$80

C \$384

D \$424

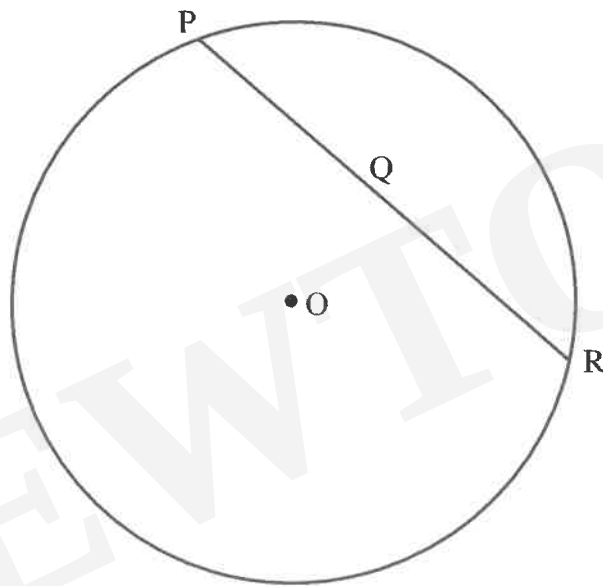
25 An angle of 95° is

- A an obtuse.
- B an acute.
- C a reflex.
- D a straight angle.

26 A man invested \$5 000 in a bank. The interest rate was 5% per annum. What would be his interest after 10 years?

- A \$250
- B \$2 500
- C \$5 250
- D \$7 500

27 In the diagram below, O is the centre of the circle.

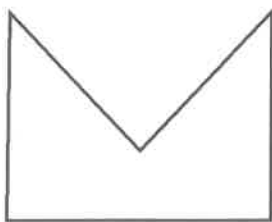


Line PQR is a

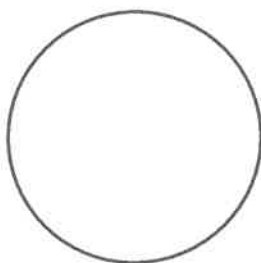
- A circumference.
- B diameter.
- C radius.
- D chord.

28 Which of the shapes below is a quadrilateral?

A



B



C



D



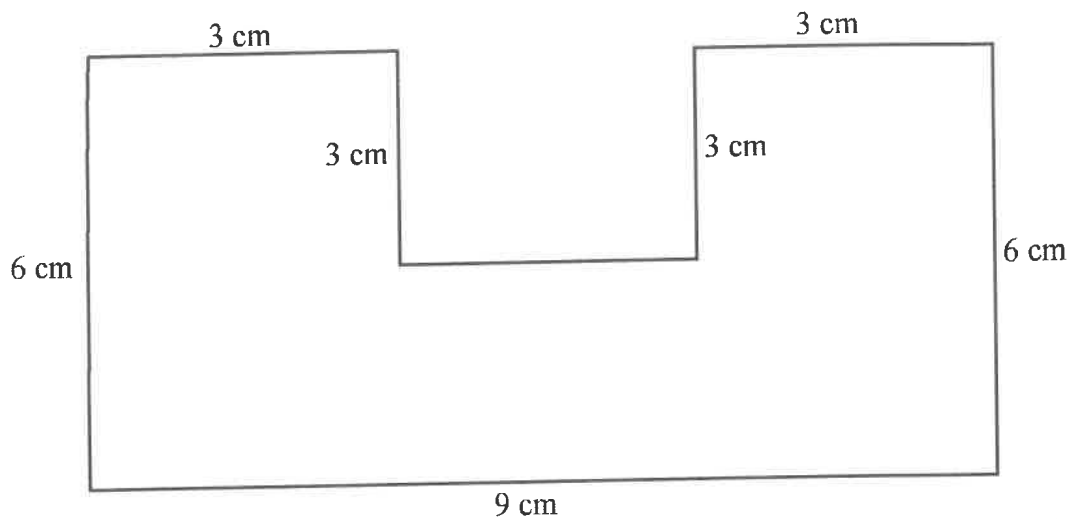
29 $\frac{3}{4}$ of 36 is

- A** 9.
- B** 12.
- C** 27.
- D** 48.

30 The sum of 12 and 0,25 is

- A** 12,25.
- B** 11,75.
- C** 1,45.
- D** 0,37.

- 31 The perimeter of the following shape is



- A 54 cm.
B 36 cm.
C 33 cm.
D 30 cm.
- 32 If 10 men take 4 days to build a house, how long will it take 8 men to build a similar house working at the same rate?
- A 5 days
B 8 days
C 10 days
D 20 days
- 33 The following table shows five students' marks in a test marked out of 50.

Student	Taku	Kudzi	Ellen	Linda	Leo
Mark	45	40	50	42	43

What was the average mark of the students?

- A 220
B 44
C 43
D 42

34 $\frac{16}{20}$ in its lowest terms is

A $\frac{8}{10}$

B $\frac{4}{10}$

C $\frac{4}{5}$

D $\frac{3}{5}$

35 The net mass of beans is 450 g. The empty tin has a mass of 115 g.
What is the gross mass of the tin of beans?

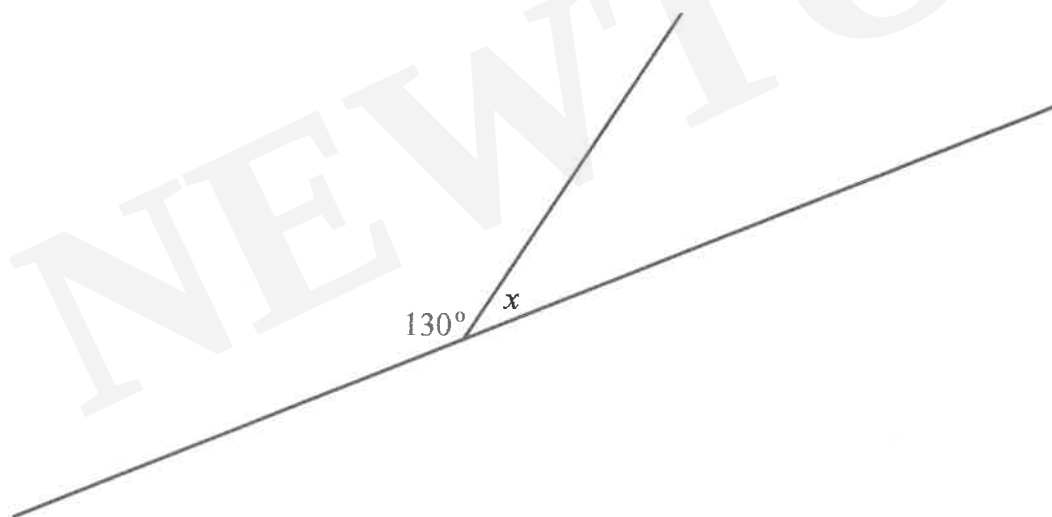
A 565 g

B 425 g

C 335 g

D 225 g

36 Calculate the angle marked x in the diagram below.



A 230°

B 180°

C 90°

D 50°

37 $1\frac{5}{8} \div 3 =$

A $3\frac{23}{25}$

B $1\frac{5}{8}$

C $\frac{13}{24}$

D $\frac{8}{29}$

38 $\frac{2}{3} + \frac{3}{5} =$

A $1\frac{4}{15}$

B $1\frac{4}{5}$

C $\frac{4}{15}$

D $\frac{5}{8}$

39 The marked price of a book was \$7,00. Takudzwa bought it at a discount of 8%.
How much did he pay for the book?

A \$7,56

B \$6,44

C \$0,56

D \$0,15

40 $(9 - 8) + 9 \times 8 =$

A 71.

B 72.

C 73.

D 89.

- 41 Eliot has \$6. He spends 75 cents.
What fraction of his money did he spend?
- A $\frac{1}{8}$
- B $\frac{3}{8}$
- C $\frac{3}{4}$
- D $\frac{2}{25}$
- 42 A sum of \$648 was shared among Peter, John and James in the ratio 2: 3: 4 respectively.
What was John's share?
- A \$288
- B \$216
- C \$144
- D \$ 72
- 43 A pond is 4 m long, 2,5 m wide and 1,5 m deep.
Calculate the volume of the water when the pond is full.
- A $15 m^3$
- B $150 m^3$
- C $1\ 500 m^3$
- D $15\ 000 m^3$
- 44 The following shows the distances in km between the major towns in Zimbabwe.

Mutare				
263	Harare			
480	280	Gweru		
577	439	164	Bulawayo	
1 309	1 046	607	443	Victoria Falls

Find the distance between Bulawayo and Mutare.

- A 263
- B 443
- C 577
- D 706

45 $3\frac{1}{2} \div \frac{1}{4} =$

A 14.

B $3\frac{3}{4}$.

C $\frac{7}{8}$.

D $\frac{4}{7}$.

46 $\frac{4}{6} - \frac{1}{3} =$

A $\frac{5}{6}$.

B $\frac{2}{3}$.

C $\frac{3}{6}$.

D $\frac{1}{3}$.

47 Study the following number sequence.

5; 6; 8; 11; 15;

What is the next number in the sequence?

A 19

B 20

C 23

D 26

48 A fuel tank of a car contains 80 l of petrol. The car uses 1 l of petrol for every 20 km travelled.

How much petrol is left in the tank if the car covers 1 000 km?

A 1 l

B 30 l

C 50 l

D 60 l

- 49 In a Grade 6 class there are 45 pupils.
If $\frac{3}{5}$ of them are boys, how many of them are girls?

A 9
B 15
C 18
D 27

- 50 Express $2\frac{3}{4}$ l in *ml*.

A 750 *ml*
B 2 725 *ml*
C 2 750 *ml*
D 2 250 *ml*

NEWTON



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
GRADE SEVEN EXAMINATION, 2019.

MATHEMATICS
PAPER 2

002/2

OCTOBER 2019 SESSION

Time: 2 hours

INSTRUCTIONS TO CANDIDATES

1. Answer **all** questions in Section **A** on the separate answer paper provided.
2. Answer any **three** questions in Section **B** on the separate answer paper provided.
3. Section **A** carries 25 marks.
4. Section **B** carries 15 marks.
5. Omission of essential working may result in loss of marks.
6. Do **not** measure from given diagrams.
7. Electronic calculators and slide rules **must not** be used in the examination.

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Section A (25 marks)

*Answer **all** questions in this section.*

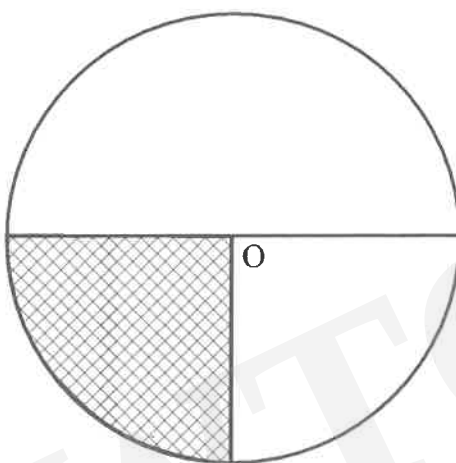
1 Write the following numbers in Arabic numerals:

(a) fifteen, [1]

(b) forty. [1]

2 The diagram below is a circle, with centre at O.

(a) What fraction of the circle is shaded? [1]



(b) Express the shaded part as a decimal fraction of the circle. [2]

3 Copy and complete the following statement.

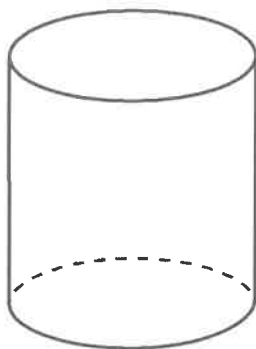
$2 + (3 + 4) = (2 + \square) + 4 = \square$ [2]

4 Find the value of

(a) $3 - 1,75$ [2]

(b) $12,4 + 2,9$ [2]

- 5 (a) State the name given to the following shape.



[1]

- (b) The volume of a cuboid is 24 cm^3 . Its length is 4 cm and its height 3 cm.

Calculate its width.

[3]

- 6 Simplify $\left(\frac{1}{2} \times 12\right) + \frac{3}{4}$.

[2]

- 7 Convert

- (a) 1 cubic metre to litres.

[1]

- (b) 20 000 m^2 to hectares.

[1]

- 8 Find the average speed, in km/h, of a car that travels 400 metres in 12 seconds.

[3]

- 9 A pole is 2 m tall. Its shadow is 3 metres long at 4 p.m.

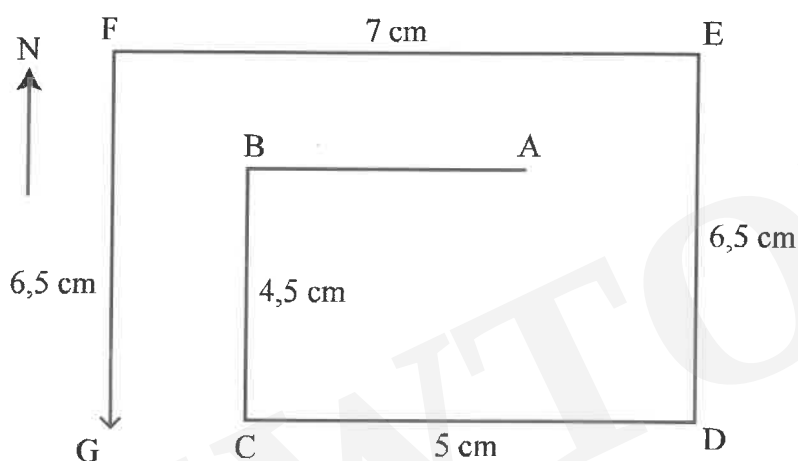
Calculate the length of a girl's shadow who is 0,8 metres tall at the same time.

[3]

Section B (15 marks)

*Answer any **three** questions from this section.*

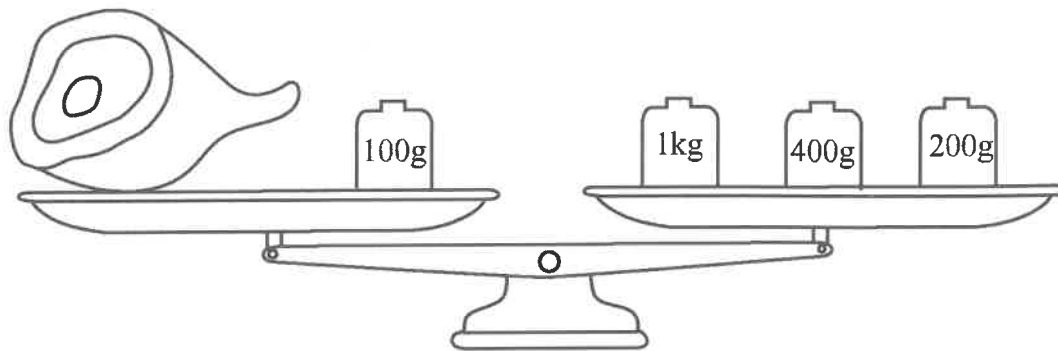
- 10** A vendor sells 24 dozens of ice-cream per day.
- (a) If she earns 5 cents commission for each ice cream sold, how much does she earn per day? [3]
- (b) If she works 7 days per week, how much does she earn per week? [2]
- 11** The following is a map showing Takunda's walk at his father's farm. Study it and answer questions that follow.



Scale: 1 cm: 1 km

- (a) In what direction was Takunda walking from A to B? [1]
- (b) Through how many right angles did he turn while walking from A to G? [1]
- (c) Calculate the actual distance that he walked from B to G. [3]

- 12** The following diagram shows a balancing scale showing meat being weighed.

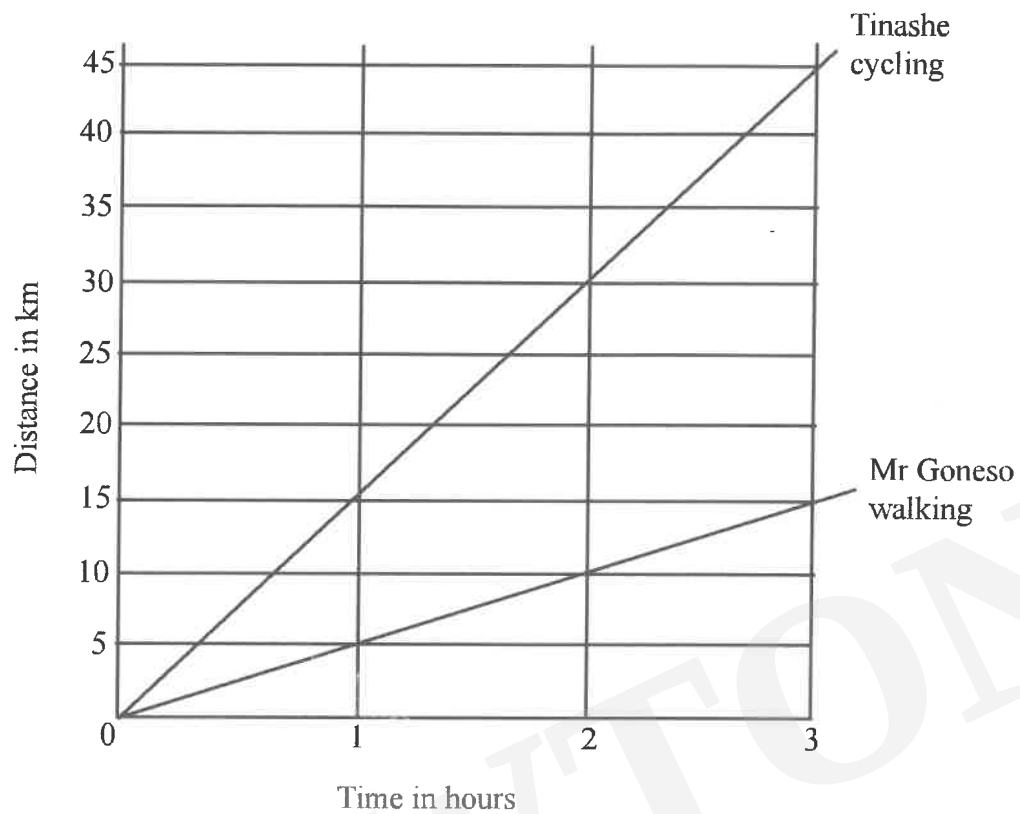


- (a) Find the weight of the meat in kilograms. [3]
- (b) If one kilogram of meat costs \$5,00, calculate the cost of the meat. [2]
- 13** Two children had notes and coins as shown in the following table.

	\$100	\$20	\$5	\$1	50c	20c
Thandi	0	0	5	4	0	3
Sipho	1	1	4	2	3	0

- (a) Find how much Thandi has in dollars. [2]
- (b) Find, in dollars, the difference between Thandi and Sipho's amounts. [3]

- 14 The following graph shows Mr Goneso's walking speed and Tinashe's cycling speed.



- (a) Who is travelling faster? [1]
- (b) Write down Mr Goneso's speed in km/h. [1]
- (c) Calculate the time taken by Tinashe to cycle 90 km, if he maintains his speed. [3]



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
GRADE SEVEN EXAMINATION, 2020.

MATHEMATICS
PAPER 1

002/1

OCTOBER 2020 SESSION

Time: 2 hours

INSTRUCTIONS TO CANDIDATES

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5. If you wish to change your answer, **erase it completely** with a pencil rubber and then shade the new choice.
6. If **more than one** box is shaded for any one answer, that answer will be regarded as **wrong**.
7. **If you do not understand** the instructions, **ask** the invigilator to explain them to you **before you start**.
8. Answer **all** the questions on the separate answer sheet provided.
9. Rough paper will be provided.

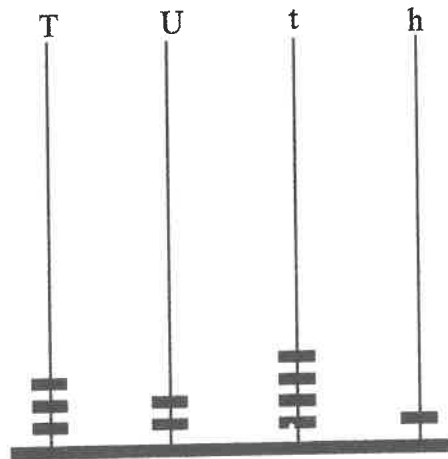
This question paper consists of 12 printed pages.

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1 Twelve comma six in numerals is

- A 1,26.
- B 12,06.
- C 12,6.
- D 126,00.

2



The number on the abacus is

- A 3,241.
- B 32,41.
- C 324,1.
- D 3 241.

3 The missing number in the pattern 2, 4, 7, 11, 16 is

- A 17.
- B 18.
- C 21.
- D 22.

4 The product of 27 and 3 is

- A 9.
- B 30.
- C 61.
- D 81.

5 Which is a square number?

- A 27
- B 44
- C 49
- D 90

- 6 Below is part of a calendar.

M	T	W	T	F	S	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30				

Which month is shown?

- A February
 B July
 C August
 D September
- 7 $\frac{2}{5}$ as a decimal is
- A 0,4.
 B 0,8.
 C 2,5.
 D 5,2.
- 8 Musa, Tendai, Siphon and Henry shared sweets in the ratio 1:2:3:4; respectively.
- Who got the largest share?
- A Siphon
 B Tendai
 C Musa
 D Henry
- 9 1 648 rounded off to the nearest 10 is
- A 1 600.
 B 1 640.
 C 1 650.
 D 2 000.
- 10 From the fractions $\frac{2}{3}$; $\frac{4}{5}$; $\frac{5}{6}$; $\frac{7}{10}$ which one is the biggest?

- A $\frac{7}{10}$
 B $\frac{5}{6}$
 C $\frac{4}{5}$
 D $\frac{2}{3}$

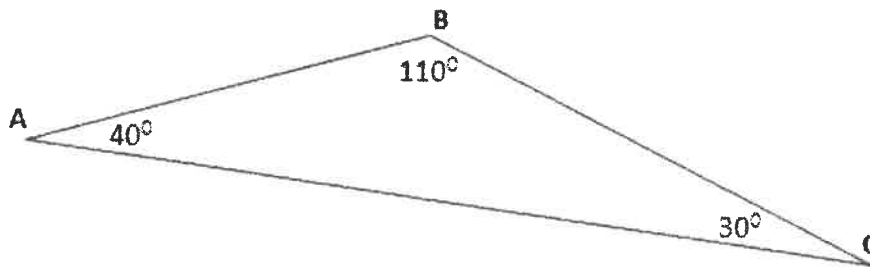
[Turn over

- 11 Simplify $\frac{4}{5} \div \frac{1}{2}$
- A $\frac{2}{5}$
B $\frac{5}{8}$
C $1\frac{3}{5}$
D $1\frac{4}{5}$
- 12 $\frac{42}{8}$ as a mixed number in its lowest terms is
- A $4\frac{1}{4}$
B $4\frac{2}{8}$
C $5\frac{2}{8}$
D $5\frac{1}{4}$
- 13 What is the difference between 900 and 237?
- A 663
B 673
C 763
D 1 137
- 14 $0,5 \times 0,5 \times 0,5 =$
- A 0,125.
B 1,250.
C 12,50.
D 125,0.
- 15 $8 - 4 \times 3 + 6 =$
- A 2.
B 18.
C 28.
D 36.
- 16 Simba got 12 out of 30 marks.
His mark expressed as a percentage is
- A 20%.
B 40%.
C 48%.
D 75%.

- 17 The perimeter of a regular pentagon with sides 6 cm is

A 30 cm.
B 36 cm.
C 42 cm.
D 48 cm.

18



In triangle ABC, the interior angle at B is

A acute.
B obtuse.
C reflex.
D complete revolution.

- 19 Below is a right-angled triangle.



The angle marked x is

A 43° .
B 47° .
C 57° .
D 133° .

[Turn over

- 20 The total of five 50 c coins, six \$5 notes and three \$2 notes is

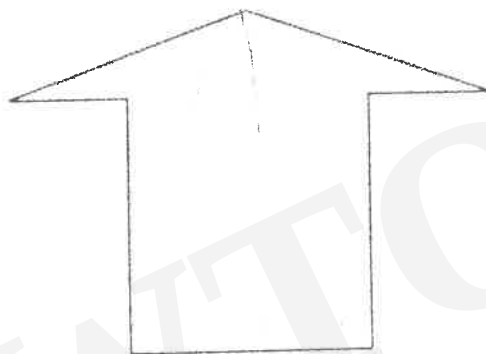
A \$7,50.
B \$11,50.
C \$38,50.
D \$61,00.

- 21 A vehicle has an empty mass of 20,5 t and a load of 37 t.

What is the gross mass?

A 16,5 t.
B 17,5 t.
C 24,2 t.
D 57,5 t.

22



The number of lines of symmetry for the shape above is

A 4.
B 3.
C 2.
D 1.

- 23 The time between 11.17 am and 1.02 pm on the same day is

A 10 h 15 min.
B 2 h 25 min.
C 1 h 45 min.
D 2 h 45 min.

- 24 Mr Jones put \$500 in a bank for $3\frac{1}{2}$ years. He was given an interest of 5% per annum.

His interest after $3\frac{1}{2}$ years was

A \$84,50.
B \$87,50.
C \$412,50.
D \$587,50.

25 The Highest Common Factor (HCF) of 24 and 36 is

- A 3.
- B 4.
- C 6.
- D 12.

26 Mr Malinga bought six suits at \$135 each. He sold the suits at \$160 each.

His profit was

- A \$25.
- B \$150.
- C \$810.
- D \$960.

27 A station sold petrol to 16 cars. Each car got 45 *litres*.

How many *litres* of petrol were sold?

- A 29 *litres*
- B 61 *litres*
- C 620 *litres*
- D 720 *litres*

28 87,5% as a fraction in its lowest terms is

- A $87\frac{1}{2}$.
- B $\frac{43}{50}$.
- C $\frac{7}{8}$.
- D $\frac{22}{25}$.

29 In January Mr Bhebhe had a balance of \$632 in his account. He deposited \$141 in February and he withdrew \$300 in March.

His new balance was

- A \$373.
- B \$473.
- C \$773.
- D \$1 073.

[Turn over



- 30 A car travelled 45 km in 25 minutes.

Its speed in km/h is

- A 540.
- B 225.
- C 108.
- D 90.

- 31 John, Peter and Chipu shared 36 apples in the ratio 2:3:4; respectively.

How many apples did Peter get?

- A 8
- B 9
- C 12
- D 16

- 32 A triangle has an area of 20 cm^2 . Its base is 8 cm.

Its height is

- A 160 cm.
- B 80 cm.
- C 10 cm.
- D 5 cm.

- 33 A discount of 10% was given on a wardrobe with a marked price of \$550.

The discount given was

- A \$55.
- B \$495.
- C \$540.
- D \$605.

- 34 5 litres of milk cost \$15,50.

The cost of 7 litres is

- A \$21,70.
- B \$77,50.
- C \$108,50.
- D \$217,00.

- 35 A number is multiplied by 3 then 6 is subtracted to get 15.

The number is

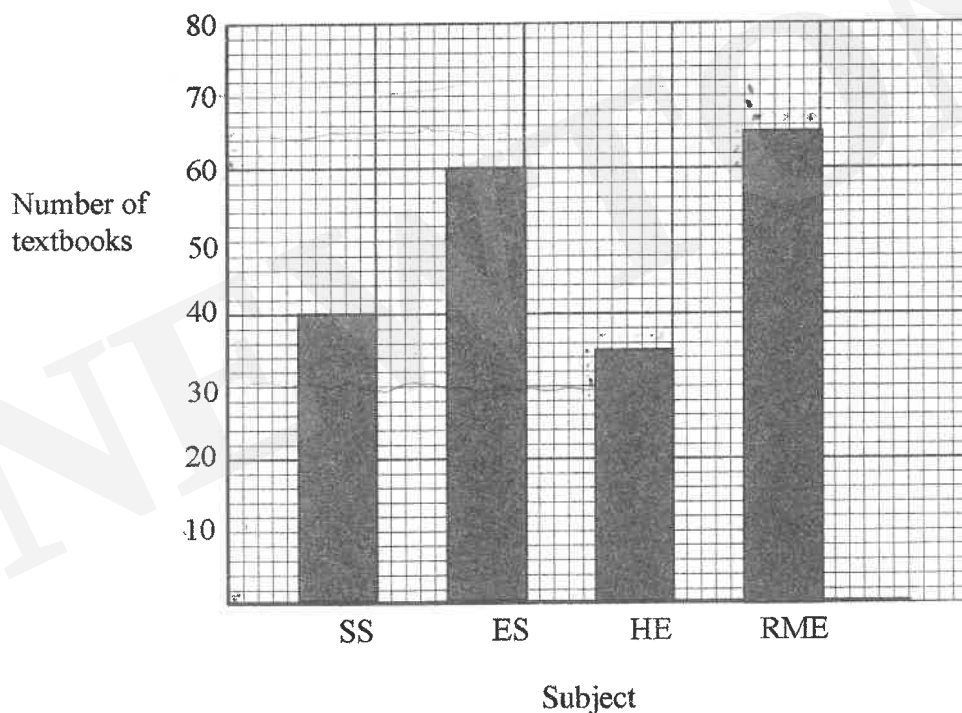
- A 21.
- B 18.
- C 9.
- D 7.

- 36 In a school of 800 learners, 95% were present.

How many learners were present?

- A 760
- B 705
- C 80
- D 40

- 37 Study the graph below.



The average number of textbooks is

- A 5.
- B 35.
- C 50.
- D 200.

- 38 $0,42 \div 0,7 =$

- A 0,006.
- B 0,06.
- C 0,6.
- D 6,0.

- 39 Five women shared $10\frac{1}{2}$ litres of cooking oil, equally.

Each woman got

- A $2\frac{1}{10}$ litres.
- B $2\frac{1}{2}$ litres.
- C $15\frac{1}{2}$ litres.
- D $50\frac{1}{10}$ litres.

- 40 The average mark for four learners is 44. Three of the learners got 45, 43 and 42.

The fourth learner got

- A 41.
- B 46.
- C 130.
- D 176.

- 41 A boy facing East turns 270° clockwise.

He is now facing

- A East.
- B North.
- C South.
- D West.

- 42 Kara left home on Tuesday the 3rd and returned the following Monday.

She returned on the

- A 9th.
- B 10th.
- C 11th.
- D 12th.

- 43 The table below shows masses of four children.

Keisha	Keith	Chipo	Busi
32 kg	35 kg	36 kg	31 kg

The difference between Keith's mass and Busi's mass is

- A 4 kg.
- B 31 kg.
- C 66 kg.
- D 134 kg.

- 44 The number of boxes measuring 10 cm by 6 cm by 4 cm that fit in a box measuring 30 cm by 18 cm by 12 cm is

A 20.
B 27.
C 240.
D 6 480.

- 45 25 000 m² expressed in hectares is

A 2 500.
B 0,25.
C 2,5.
D 25.

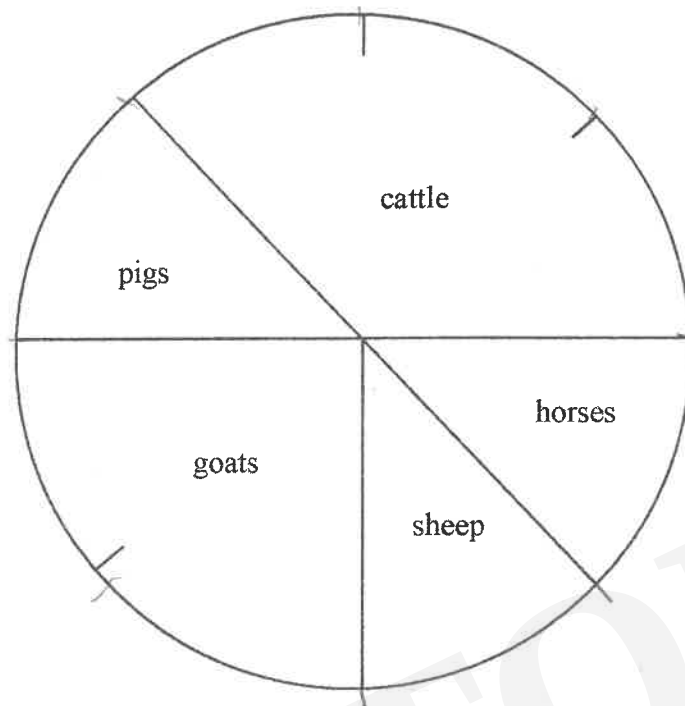
- 46 $\frac{1}{2}$ of the class walk to school, $\frac{1}{3}$ travel by bus and the rest travel by car.

What fraction of the class travel by car?

A $\frac{1}{6}$
B $\frac{2}{5}$
C $\frac{3}{5}$
D $\frac{5}{6}$

The pie chart below shows animals kept on a farm.

Use the pie chart to answer question 47 and 48.



- 47 The total number of goats and sheep as a decimal fraction of all animals is
- A 0,125.
 - B 0,375.
 - C 0,25.
 - D 0,75.
- 48 If the total number of cattle and horses is 125, then the total number of the animals at the farm is
- A 125
 - B 250
 - C 375
 - D 675
- 49 The unit used to measure capacity is
- A *centimetres.*
 - B *kilograms.*
 - C *litres.*
 - D *kilometres.*
- 50 The total number of days for June, July, August and October is
- A 120.
 - B 121.
 - C 122.
 - D 123.



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
GRADE SEVEN EXAMINATION, 2020.

MATHEMATICS
PAPER 2

002/2

OCTOBER 2020 SESSION

Time: 2 hours

Additional material:
Answer paper

INSTRUCTIONS TO CANDIDATES

1. Answer **all** questions in Section **A**.
2. Answer any **three** questions in Section **B**.
3. Section **A** carries 25 marks.
4. Section **B** carries 15 marks.
5. Omission of essential working may result in loss of marks.
6. Do **not** measure from given diagrams.
7. Electronic calculators and slide rules **must not** be used in the examination.

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ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
GRADE SEVEN EXAMINATION, 2020.

MATHEMATICS
PAPER 2

002/2

OCTOBER 2020 SESSION

Time: 2 hours

Additional material:
Answer paper

INSTRUCTIONS TO CANDIDATES

1. Answer **all** questions in Section **A**.
2. Answer any **three** questions in Section **B**.
3. Section **A** carries 25 marks.
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Section A (25 marks)

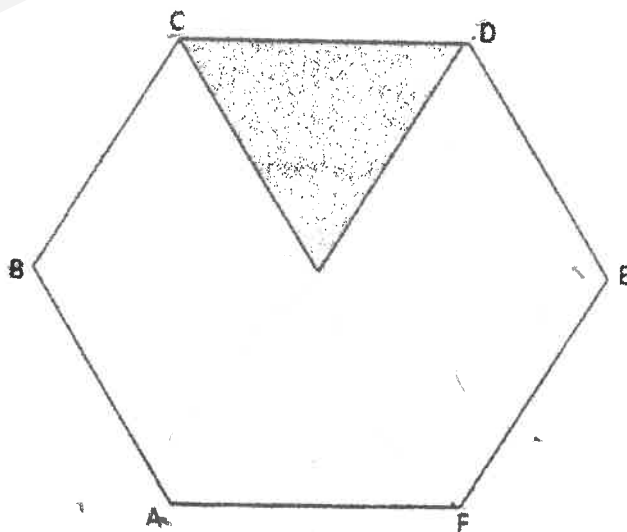
Answer *all* questions in this section.

1. (a) Write 15,43 in words. [1]
- (b) Write 499 in Roman numerals. [1]
2. Use the abacus below to answer the question that follow.

TTH	TH	H	T	U
•••		••• ••• •••	••	••• ••• •••
••• ••• •••	••• ••• •••		•• ••• •••	•

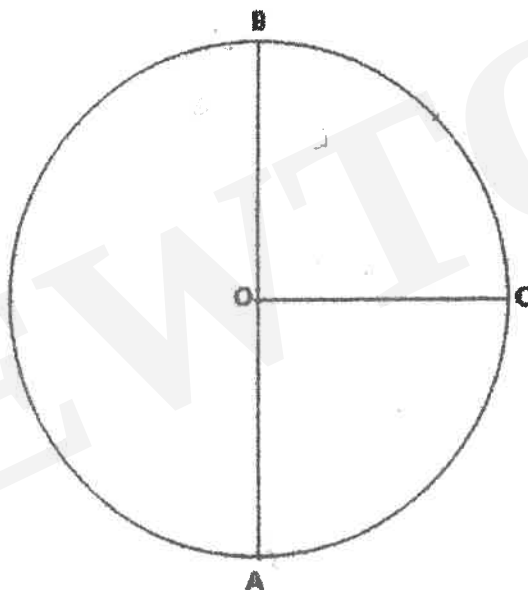
State the number shown on the abacus.

3. Simplify (a) $3 - 0,04$ [1]
- (b) $9^2 \div 10^0$ [2]
4. (a) Below is a regular shape, ABCDEF, with part of it shaded. [2]



- (i) Give the special name of the shape ABCDEF. [1]
- (ii) If the area of the shaded part is $4,5 \text{ cm}^2$, find the area of the whole shape. [2]

- (b) The perimeter of a rectangle is 24 cm. Its length is 7 cm.
Find its width. [3]
5. (a) Express 0,85 as a fraction in its lowest terms. [2]
- (b) Simplify $\frac{1}{3} \times 6 + \frac{3}{4} \div \frac{1}{2}$. [3]
6. Thandi started her journey at 9.15 p.m. and arrived at 6.15 a.m. the following day.
Calculate the time she took to complete the journey, in hours. [3]
7. (a) Express 18 April 1980 in SI notation. [1]
- (b) Below is a circle with centre at O. Points A, B and C are on the circumference of the circle.

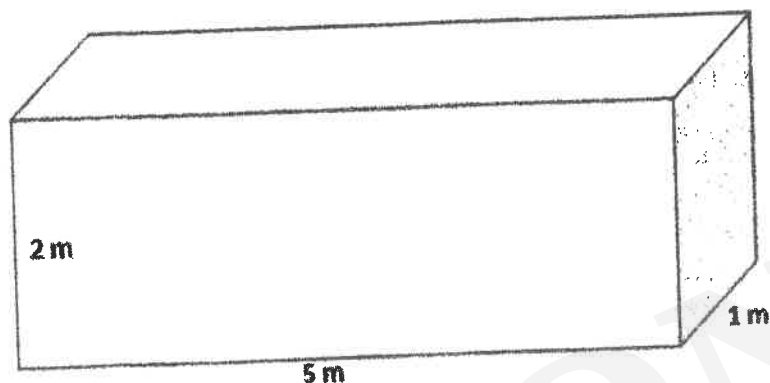


- (i) If straight line AB is 11 cm, what is the length of straight line OC? [1]
- (ii) If arc AB is 18 cm, find the length of arc BC. [2]

Section B (15 marks)

There are five questions in this section. Answer any three.

8. Below is a diagram of a water tank measuring 5 m by 2 m by 1 m.



- (a) Calculate the volume of the water in cm^3 , when the tank is full. [2]
- (b) Find the number of 20 litre buckets that can be filled from the tank when it is full. [3]
- 9 For people attending a party, $\frac{1}{3}$ of them are women, $\frac{2}{5}$ are men and the rest are children.
- (a) Express the fraction representing men as a decimal. [1]
- (b) Compare the fraction of men to the fraction of women using $>$, $<$ or $=$. [1]
- (c) Calculate the fraction representing children at the party. [3]

10. Study the fare table for Chiro Bus Company. The fares are charged at 20 cents per kilometre.

Mutare			
\$18,60	Rusape		
\$26,20	\$9,00	Macheke	
\$38,20	\$19,00	\$12,80	Marondera
\$52,60	\$34,00	\$26,40	\$14,80 Harare

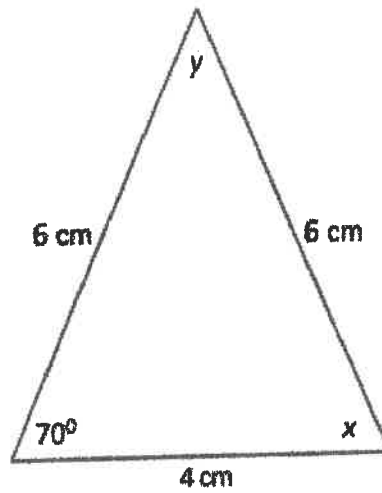
- (a) Which route on the table has the highest fare? [1]
- (b) Mr Peters travelled from Marondera to Macheke on Monday and continued to Mutare on Tuesday.
How much did he pay altogether? [2]
- (c) Mrs Moyo travelled from Rusape to Harare.
Calculate the distance she travelled. [2]

11. Tula and Noma contributed \$200 and \$300; respectively to start a business.

- (a) Express Tula's share as a fraction of Noma's share, in its lowest terms. [1]
- (b) Express Noma's contribution to Tula's contribution as a ratio in its lowest terms. [2]
- (c) They made a profit of \$120 and shared it in the ratio of their contributions.
Calculate Tula's profit. [2]

371 316 777 44

12. Study the diagram below and answer the questions that follow.



- (a) Give a special name for the triangle above. [1]
- (b) Write down the size of the angle marked x . [1]
- (c) Calculate the angle marked y . [3]



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
GRADE SEVEN EXAMINATION, 2021.

MATHEMATICS
PAPER 2

7020/02

OCTOBER 2021 SESSION

Time: 2 hours

Candidates answer on the question paper.

INSTRUCTIONS TO CANDIDATES

1. Answer **all** questions in Section A and any **three** questions in Section B.
2. Write your answers on the spaces provided below each question.
3. Necessary working must be clearly shown in the blank spaces below each question.
4. Section A carries 25 marks and Section B carries 15 marks.
5. If more than three questions are answered in Section B, only the first three will be considered.
6. Do **not** measure from given diagrams.
7. Electronic calculators and slide rules **must not** be used in the examination.

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ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
GRADE SEVEN EXAMINATION, 2021.

MATHEMATICS
PAPER 2

7020/02

OCTOBER 2021 SESSION

Time: 2 hours

Candidates answer on the question paper.

INSTRUCTIONS TO CANDIDATES

1. Answer **all** questions in Section A and any **three** questions in Section B.
2. Write your answers on the spaces provided below each question.
3. Necessary working must be clearly shown in the blank spaces below each question.
4. Section A carries 25 marks and Section B carries 15 marks.
5. If more than three questions are answered in Section B, only the first three will be considered.
6. Do **not** measure from given diagrams.
7. Electronic calculators and slide rules **must not** be used in the examination.

This question paper consists of 9 printed pages and 3 blank pages.

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Section A (25 marks)

Answer *all* questions in this section.

- 1 (a) Write 54 in Roman numerals.
(b) Express 36 as a product of its prime factors.

Answer: (a) _____ [1]
(b) _____ [2]

- 2 (a) Add $3,15 + 1,02$.
(b) Simplify $\frac{5}{8} - \frac{1}{3}$.

Answer: (a) _____ [1]
(b) _____ [2]

- 3 (a) Round off 72,651 to the nearest whole number.
- (b) Express $\frac{3}{8}$ as a percentage.

Answer: (a) _____ [1]

(b) _____ [2]

- 4 Study the invoice below.

Dingani Shop		
Items	\$	C
Cooking oil	60	00
Flour	55	00
Rice	82	70
Total	—	—

Calculate the total cost of the items.

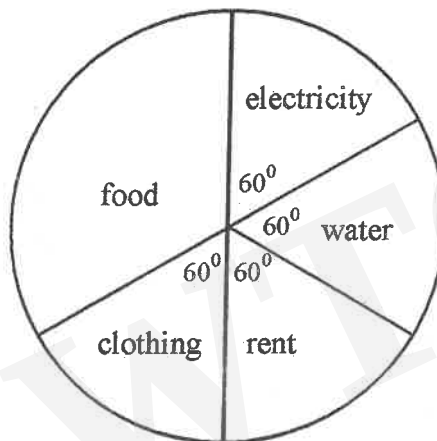
Answer: _____ [2]

[Turn over

- 5 Simplify $24 - 6 \div 3$.

Answer: _____ [2]

- 6 The diagram below shows how Mrs Dube spent her salary of \$3 000.



- (a) On which item did Mrs Dube spend more money?
- (b) Calculate the difference between the amount she spent on food and on clothing.

Answer: (a) _____ [1]

(b) _____ [4]

- 7 A bus travels at an average speed of 100 km/h.

Calculate the time it takes to cover 500 km.

Answer: _____ [2]

- 8 A farmer bought 20 bags of fertiliser each with a mass of 50 kg.
They were put in a lorry with a net mass of 3 000 kg.

- (a) State the mass of one bag of fertiliser, in grammes.
(b) Calculate the mass of 20 bags of fertiliser.
(c) Find the gross mass of the lorry, in kilograms.

Answer: (a) _____ [1]
(b) _____ [2]
(c) _____ [2]

[Turn over

Section B (15 marks)

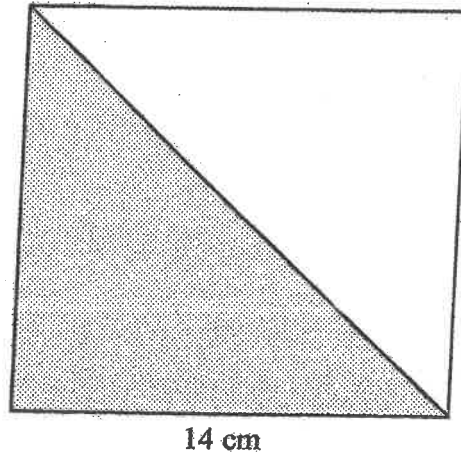
Answer any three questions from this section.

- 9 17 out of 51 learners in Grade 7 are computer literate.
- (a) Write the number of learners who are computer literate as a fraction of all learners in Grade 7, in its lowest terms.
- (b) Find $\frac{1}{2}$ of those who are not computer literate.

Answer: (a) _____ [2]

(b) _____ [3]

- 10 (a) State the number of lines of symmetry of a square.
- (b) The shape below is a square.



Calculate the

- (i) area of the shaded part.
- (ii) perimeter of the square.

Answer: (a) _____ [1]

(b)(i) _____ [2]

(ii) _____ [2]

[Turn over

11 Ben, Lindiwe and Themba shared \$3 600 in the ratio 2: 3: 4, respectively.

- (a) State the ratio of Themba to Lindiwe's share.
- (b) Calculate Lindiwe's share.
- (c) Find the total amount for Ben and Themba.

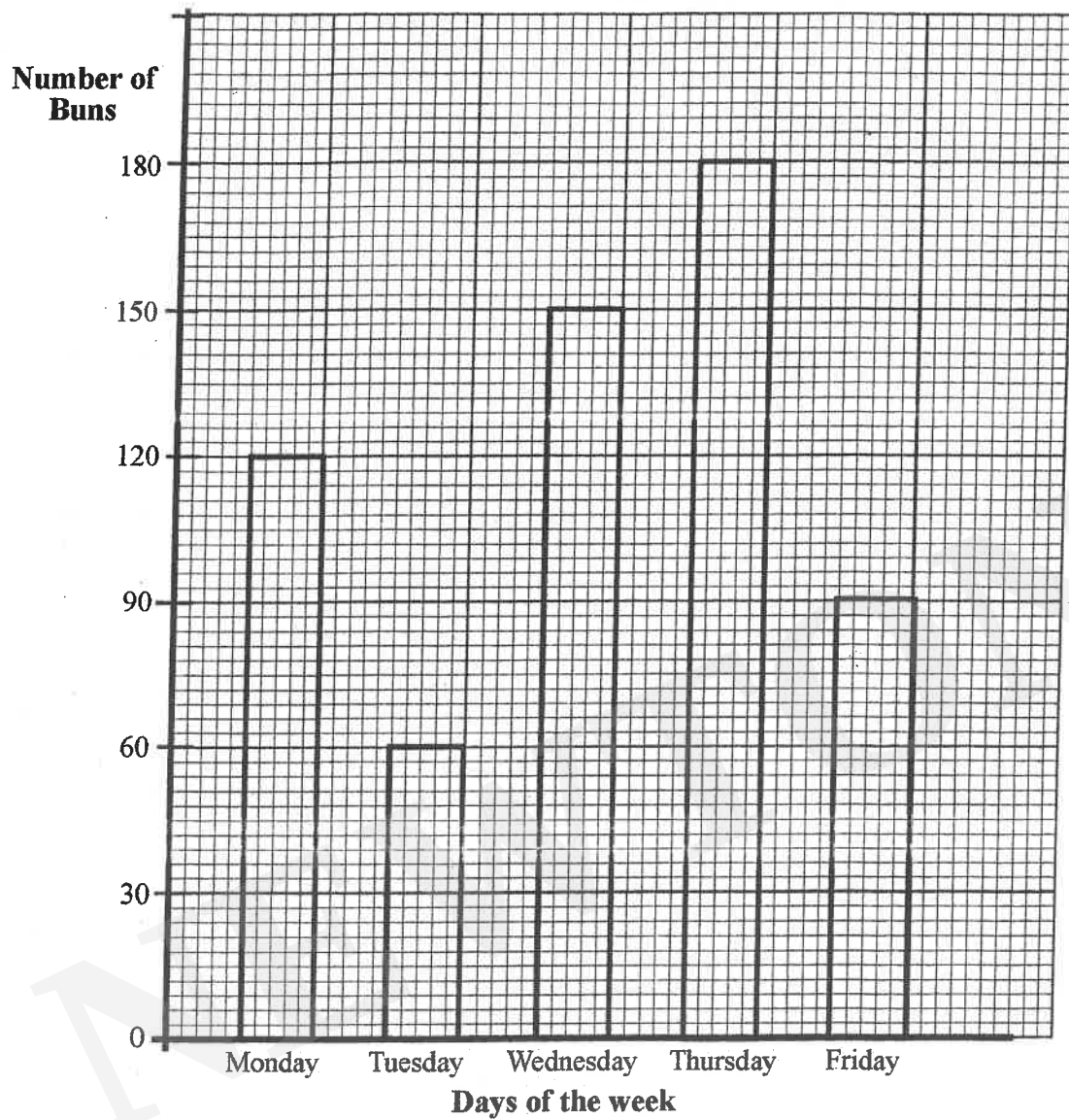
Answer:

(a) _____ [1]

(b) _____ [2]

(c) _____ [2]

- 12 The graph below shows the number of buns sold by a shopkeeper from Monday to Friday.



- (a) Find the number of buns sold in the 5 days.
- (b) Calculate the total amount the shopkeeper got in the 5 days.

Give your answer in dollars.

Answer: (a) _____ [2]

(b) _____ [3]



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
GRADE SEVEN EXAMINATION, 2021.

MATHEMATICS
PAPER 1

7020/01

OCTOBER 2021 SESSION

Time: 2 hours

INSTRUCTIONS TO CANDIDATES

1. Read all the instructions carefully.
2. Do not open this booklet until you are told to do so by the invigilator.
3. Use only an HB pencil for all entries on the answer sheet.
4. When you are told to start, choose **one** correct answer from the suggested answers and shade it **very dark** as shown in the examples at the top of the answer sheet.
5. If you wish to change your answer, erase it **completely** with a pencil rubber and then shade the new choice.
6. If more than one box is shaded for any one answer, that answer will be regarded as **wrong**.
7. If you do not understand the instructions, ask the invigilator to explain them to you **before you start**.
8. Answer **all** the questions on the separate answer sheet provided.
9. Paper for rough work will be provided.

This question paper consists of 11 printed pages and 1 blank page.

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1 Four hundred and eight in figures is

- A 408.
- B 480.
- C 4 008.
- D 4 800.

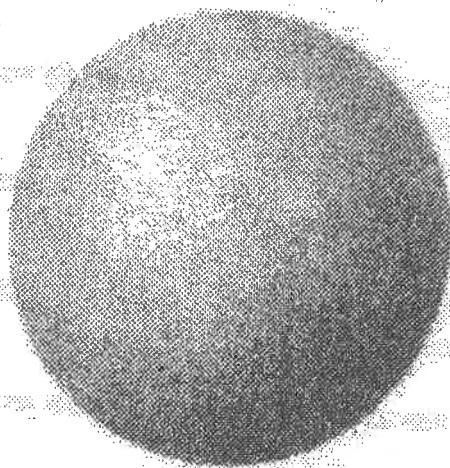
2 8 count on 4 is

- A 2.
- B 4.
- C 12.
- D 32.

3 $3\frac{3}{8}$ as an improper fraction is

- A $\frac{14}{8}$.
- B $\frac{17}{8}$.
- C $\frac{21}{8}$.
- D $\frac{27}{8}$.

4



The shape given above is a

- A cone.
- B semi-circle.
- C pyramid.
- D sphere.

5 The formula for calculating volume of a rectangular prism is

- A $2(L + W)$
- B $L \times W$
- C $L \times W \times H$
- D $L + W + H$

6 The value of 9 in 14 956 is

- A 9 000.
- B 900.
- C 90.
- D 9.

7 Which among these was a leap year?

- A 2003
- B 2005
- C 2008
- D 2010

8 If $25 + 15 = X + 25$, then $X =$

- A 10.
- B 15.
- C 25.
- D 40.

9 What is the value of four \$5 notes, two \$1 coins, two \$0,50 coins and one \$0,25 coin?

- A \$20,25
- B \$22,75
- C \$23,25
- D \$23,35

10 The table shows the number of learners, by gender, who took part in a clean up campaign at Masomera Business Centre.

Gender	Boys	Girls
Number	85	101

How many learners took part in the clean up campaign altogether?

- A 16
- B 85
- C 101
- D 186

- 11 The time shown on the clockface is 1025.

In 1 hour 15 minutes the time will be

- A 1140
- B 1130
- C 0930
- D 0910

- 12 Praise bought 10 cellphones for \$680. He sold each cellphone at \$100.

His profit was

- A \$320.
- B \$420.
- C \$580.
- D \$780.

- 13 ²⁰⁰⁰⁰515 216 rounded off to the nearest 10 000 is

- A 500 000.
- B 510 000.
- C 520 000.
- D 600 000.

- 14 The product of two numbers is 36. Their difference is 5.

The numbers are

- A 1 and 36.
- B 2 and 18.
- C 4 and 9.
- D 6 and 6.

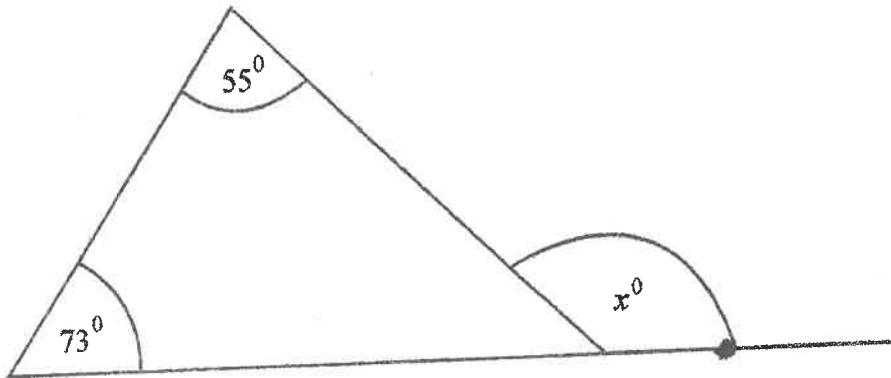
- 15 $\frac{3}{4}$ is the same as

- A $\frac{6}{10}$.
- B $\frac{12}{16}$.
- C $\frac{16}{20}$.
- D $\frac{20}{28}$.

- 16 If the perimeter of a square is 36 cm, then its area is

A 9 cm².
 B 18 cm².
 C 72 cm².
 D 81 cm².

17



The angle marked x is

A 128° .
 B 125° .
 C 107° .
 D 52° .

- 18 The table shows daily newspaper sales.

Day	Sun	Mon	Tue	Wed	Thu	Fri	Sat
Sales	363	249	100	79	88	106	264

Which day had the least sales?

A Sunday
 B Thursday
 C Saturday
 D Wednesday

- 19 $\frac{7}{8}$ expressed as a percentage is

A 74.5.
 B 77.5.
 C 84.5.
 D 87.5.

- 20 Mrs Musaka had \$6 000 in her bank account. She withdrew \$2 500.

Her new balance is

- A \$3 500.
- B \$4 000.
- C \$4 500.
- D \$8 500.

- 21 Chester Primary School bought 50 laptops for \$100 000.

The cost of each laptop is

- A \$200 000.
- B \$20 000.
- C \$2 000.
- D \$200.

- 22 If $\frac{3}{4}$ of a number is 24, then the number is

- A 6.
- B 8.
- C 18.
- D 32.

- 23 In an orchard there are 42 orange trees in a row.

How many orange trees are there in 38 rows?

- A 1 596
- B 1 586
- C 462
- D 80

- 24 7.23 rounded off to the nearest tenth is

- A 7.
- B 8.
- C 7.2.
- D 7.3.

- 25 Anna ran 100 m in 10 seconds.

Her speed in km/h was

- A 6.
- B 10.
- C 36.
- D 360.

- 26 Karabo bought six dozens of eggs.

How many eggs did she buy altogether?

- A 18
- B 60
- C 62
- D 72

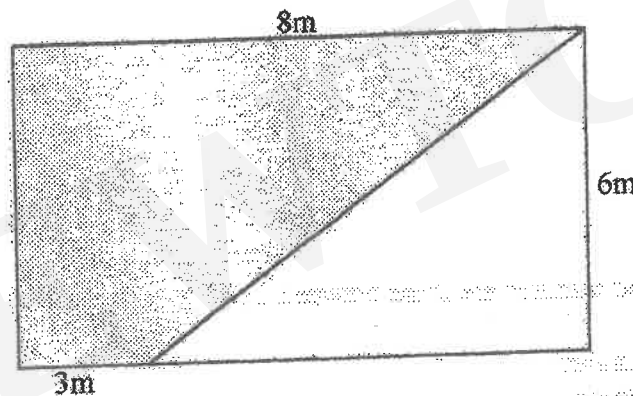
- 27 The table shows the temperature in Maphisa for the first five days of September.

Day	Mon	Tue	Wed	Thu	Fri
Temperature ($^{\circ}\text{C}$)	31	32	30	32	35

The average temperature in five days is

- A 30°C .
- B 32°C .
- C 150°C .
- D 160°C .

- 28



The diagram shows a rectangle with a part of it shaded.

The area of the unshaded part is

- A 15 m^2 .
- B 24 m^2 .
- C 30 m^2 .
- D 33 m^2 .

- 29 Victor and Themba shared \$100 such that Themba got \$20 more than Victor.

Victor got

- A \$30.
- B \$40.
- C \$60.
- D \$80.

[Turn over

- 30 A school has 1 000 learners.

If 5% of the learners were absent on a particular day, how many learners were present?

- A 10
- B 50
- C 200
- D 950

- 31 $6 \div 3 \times 2 + 4 =$

- A 8.
- B 5.
- C 3.
- D 1.

- 32 How many 6 cm by 4 cm tiles can fit into a rectangle measuring 12 cm by 12 cm?

- A 6
- B 8
- C 24
- D 144

- 33 $2,4 \div 0,6 =$

- A 40
- B 4
- C 0,4
- D 0,04

- 34 How many hours and minutes are there between 11.15 a.m. and 12.30 p.m. of the same day?

- A 1 hour 10 minutes
- B 1 hour 15 minutes
- C 1 hour 25 minutes
- D 1 hour 30 minutes

- 35 Lindani bought a Wifi router marked \$400 under hire purchase terms. She paid a deposit of 25% and was charged 10% interest on the balance.

The interest charged was

- A \$30.
- B \$75.
- C \$100.
- D \$300.

36 $\frac{18}{24}$ reduced to its lowest terms is

A $\frac{3}{4}$

B $\frac{2}{3}$

C $\frac{6}{8}$

D $\frac{9}{12}$

37 Kundai invested \$140 000 at 10% interest per annum for $2\frac{1}{2}$ years.

Her interest was

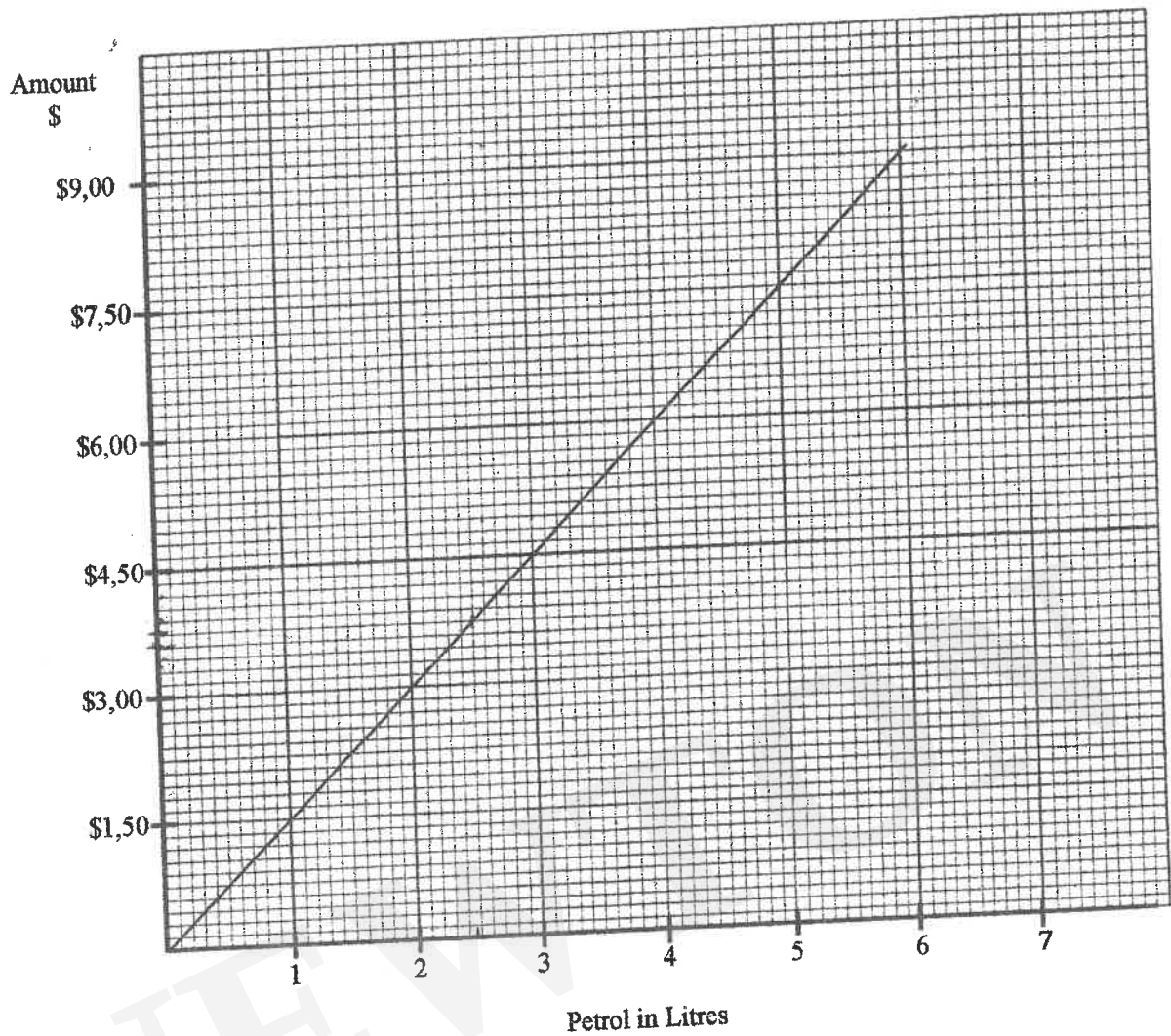
A \$5 600.

B \$7 000.

C \$14 000.

D \$35 000.

- 38 The diagram shows the cost of petrol at a filling station.

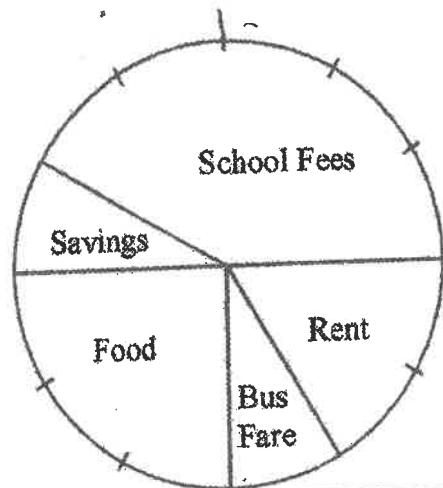


What is the cost of $2\frac{1}{2}$ litres of petrol?

- A \$0,50
- B \$3,50
- C \$3,75
- D \$4,50

Use the pie chart to answer questions 39 and 40.

The pie chart shows how Mrs Ncube spent her salary of \$500.



39 How much did she spend on food?

- A \$125
- B \$123
- C \$120
- D \$100

40 The fraction of salary spent on rent, expressed as a proper fraction in its lowest terms is

- A $\frac{3}{12}$
- B $\frac{2}{12}$
- C $\frac{1}{6}$
- D $\frac{1}{4}$



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

GRADE SEVEN EXAMINATION, 2022.

MATHEMATICS
PAPER 1

7020/1

OCTOBER 2022 SESSION

Time: 2 hours

Electronic calculators must not be used.

INSTRUCTIONS TO CANDIDATES

1. Read **all** the instructions carefully.
2. **Do not** open this booklet until you are told to do so by the invigilator.
3. Use only an HB pencil for all entries on the answer sheet.
4. When you are told to start, choose **one** correct answer from the suggested answers and shade it **very dark** as shown in the examples at the top of the answer sheet.
5. If you wish to change your answer, **erase it completely** with a pencil rubber and then shade the new choice.
6. If **more than one** box is shaded for any one answer, that answer will be regarded as **wrong**.
7. **If you do not understand** the instructions, **ask** the invigilator to **explain** them to you **before you start**.
8. Answer **all** the questions on the separate answer sheet provided.
9. Paper for rough work will be provided.

This question paper consists of 10 printed pages and 2 blank pages.

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- 1 6 002 in words is
- A sixty and zero two.
 - B six zero zero and two.
 - C six hundred and two.
 - D six thousand and two.
- 2 10 count back 2 is
- A 5.
 - B 8.
 - C 12.
 - D 20.
- 3 Add 538 and 63.
- A 601
 - B 591
 - C 535
 - D 475
- 4 Which of the following is a prime number?
- A 2
 - B 9
 - C 15
 - D 18
- 5 3 078 rounded off to the nearest ten is
- A 3070.
 - B 3080.
 - C 3088.
 - D 3100.
- 6 $2,03 + 0,3 + 3 =$
- A 2,09.
 - B 2,63.
 - C 5,06.
 - D 5,33.



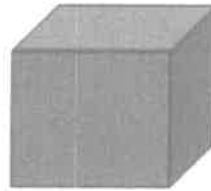
- 7 From the shapes below which one is 2 dimensional?



A



B



C



D

- 8 1299; 1219; 1289; 1291 arranged in ascending order is

- A 1299; 1291; 1289; 1219.
- B 1289; 1299; 1291; 1219.
- C 1219; 1289; 1291; 1299.
- D 1299; 1289; 1219; 1291.

- 9 $\frac{5}{8}$ as a decimal is

- A 0,625.
- B 0,725.
- C 6,250.
- D 8,250.

- 10 If $\frac{2}{3}$ of a number is 24, the number is

- A 8.
- B 12.
- C 16.
- D 36.

- 11 Seven litres of petrol cost \$81,55.

The cost of a litre of petrol is

- A \$10,22.
- B \$11,65.
- C \$11,66.
- D \$11,85.

- 12 Three minutes past twelve midnight in 24 hour notation is

- A 0003.
- B 0300.
- C 0312.
- D 1203.

[Turn over



- 13 Chido was facing East. She turned through three right angles anticlockwise.

Now she is facing

- A East.
- B North.
- C South.
- D West.

The table below shows the timetable of a bus travelling from Masvingo to Harare.

Use it to answer questions 14 and 15.

Station	Time
Masvingo	- - Dep 0900
Mvuma	Arr 1015 Dep 1020
Chivhu	Arr 1105 Dep 1115
Beatrice	Arr 1210 Dep 1220
Harare	Arr 1250 - -

- 14 How long does the journey from Mvuma to Chivhu take?

- A 45 minutes
- B 50 minutes
- C 55 minutes
- D 60 minutes

- 15 The time taken to cover the whole journey including stoppage time is

- A 2 hours 25 minutes.
- B 2 hours 35 minutes.
- C 3 hours 25 minutes.
- D 3 hours 50 minutes.

- 16 Keisha's span is 12 cm. She measured the width of a window and found 13 spans.

The width of the window is

- A 13,00 m.
- B 12,00 m.
- C 1,56 m
- D 1,46 m



17 $\frac{8}{40}$ in its lowest terms is

A $\frac{8}{20}$

B $\frac{2}{10}$

C $\frac{1}{5}$

D $\frac{1}{4}$

18 The perimeter of a square measuring 7 cm is

A 14 cm.

B 28 cm.

C 49 cm.

D 56 cm.

19 Ruth invested \$300 at 6% per annum for 3 years.

Her interest was

A \$18.

B \$34.

C \$50.

D \$54.

The table below shows attendance of 60 learners in a Grade Seven class at Mukotosi School. Use it to answer questions 20 and 21.

Days	Monday	Tuesday	Wednesday	Thursday	Friday
Number of learners	60	58	60	55	57

20 The day that had the least number of learners is

A Wednesday.

B Thursday.

C Tuesday.

D Friday.

21 The average attendance of the learners was

A 55.

B 57.

C 58.

D 60.

[Turn over



- 22 Langa took 5 ml of honey syrup three times a day for five days from a 100 ml bottle.

The remaining honey syrup was

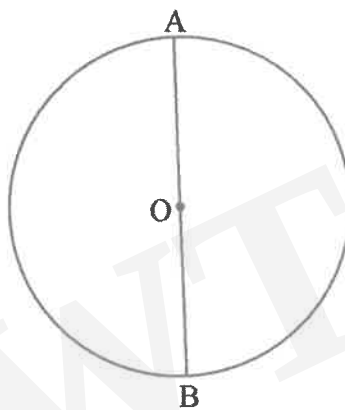
- A 75 ml.
- B 25 ml.
- C 20 ml.
- D 15 ml.

- 23 Sam paid \$28 towards water charges. He paid with two \$20 notes.

His change was

- A \$12.
- B \$22.
- C \$28.
- D \$68.

- 24 The diagram below is a circle with centre O. Line AB is the diameter measuring 7 cm.



The radius is

- A 3,1 cm.
- B 3,5 cm.
- C 7 cm.
- D 14 cm.



Below is a calendar showing the month of February. Use it to answer questions 25 and 26.

M	T	W	T	F	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28			

25 On which day was the 31st of January?

- A Wednesday
- B Thursday
- C Monday
- D Sunday

26 James worked from Monday to Thursday.

How many days did he work in February?

- A 16
- B 20
- C 25
- D 28

27 The gross mass of a tin of jam is 750 g and the mass of the empty container is 35 g.

The net mass is

- A 785 g.
- B 725 g.
- C 715 g
- D 625 g.

28 A car travelled a distance of 416 km in 4 hours.

Its average speed was

- A 14 km/h.
- B 28 km/h.
- C 104 km/h.
- D 208 km/h.

[Turn over

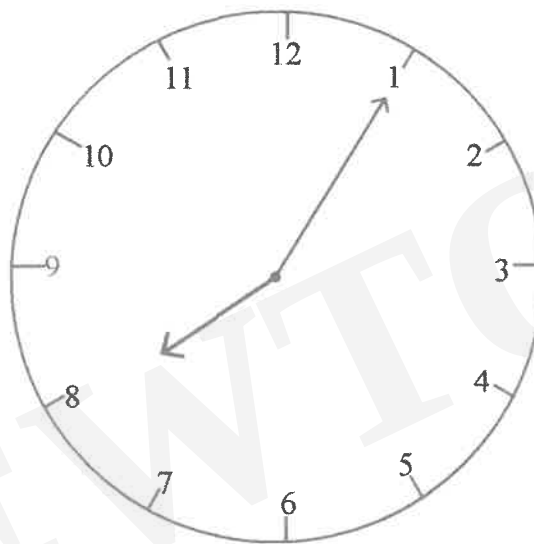


- 29 A vendor bought 963 oranges and later sold $\frac{4}{9}$ of them.

How many oranges were left?

- A 428
 - B 535
 - C 545
 - D 895
- 30 $\frac{2}{5}$ of \$1 is
- A 4 c.
 - B 20 c.
 - C 22 c.
 - D 40 c.

31



If the clock is 15 minutes behind, the correct time is

- A 10 minutes to 8.
- B 10 minutes past 8.
- C 20 minutes past 8.
- D 5 minutes past 8.



The table below shows daily temperature recorded from Sunday to Thursday of a certain week.

Day	Sun	Mon	Tue	Wed	Thur
Temperature	25°C	26°C	21°C	23°C	-

Use the table to answer questions 32 and 33.

- 32 The average temperature for the five days is 23°C.

What was the temperature recorded on Thursday?

- A 24 °C
- B 22 °C
- C 21 °C
- D 20 °C

- 33 The total temperature for the first three days was

- A 64 °C.
- B 70 °C.
- C 72 °C.
- D 74 °C.

- 34 Mrs Moyo had \$284,50 in her bank account. She withdrew \$150,00 on Monday and then deposited \$375,00 on Wednesday.

Her new balance is

- A \$434,50.
- B \$509,50.
- C \$659,50.
- D \$809,50.

- 35 $0,3 \times 0,3 \times 0,3 =$

- A 0,9.
- B 0,09.
- C 0,27.
- D 0,027.

- 36 Sibho purchased 112 seedlings at \$8 each.

She paid

- A \$896.
- B \$886.
- C \$120.
- D \$104.

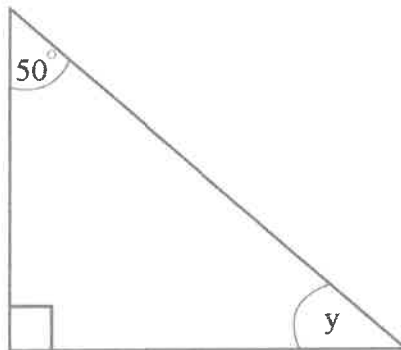
[Turn over



37 Three eights as a percentage is

- A 36,2.
- B 36,5.
- C 37,2.
- D 37,5.

38 The diagram below shows a right-angled triangle with angles as shown.



Angle y is

- A 40° .
- B 50° .
- C 130° .
- D 140° .

39 The length of a rectangular prism with a volume of 192 m^3 , width of 6 m and a height of 4 m is

- A 4 m.
- B 8 m.
- C 24 m.
- D 48 m.

40 A class has 42 learners. One third of them are boys.

How many are girls?

- A 14
- B 21
- C 28
- D 42



Candidate Name

Centre Number

Candidate Number



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

GRADE SEVEN EXAMINATION, 2022.

MATHEMATICS PAPER 2

7020/2

LANGUAGE MODIFIED PAPER FOR CANDIDATES WITH HEARING IMPAIRMENT

OCTOBER 2022 SESSION

Time: 2 hours

Do not use Electronic calculators.

INSTRUCTIONS TO CANDIDATES

1. Write your name, centre number and candidate number on the spaces at the top of each page.
2. Answer **all** questions in Section A and **three** questions in Section B.
3. If more than **three** questions are answered in **Section B**, the first three will be considered.
4. Show your working in the space below the question.
5. Write your answers in the spaces indicated below each question.
6. You lose marks for leaving out essential working.
7. Do **not** measure from given diagrams.
8. Do not use Electronic calculators and slide rules.

FOR EXAMINER'S USE

Section A

1

2

3

4

5

6

7

Section B

TOTAL

This question paper consists of 10 printed pages and 2 blank pages.

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Candidate Name

Centre Number

Candidate Number

Section A (25 marks)

Answer all questions in this section.

- 1 (a) Write $\frac{3}{5}$ in words.
- (b) Express 599 in Roman numerals.

Answer (a) _____ [1]

(b) _____ [1]

- 2 (a) Write 84 016 in expanded index notation.
- (b) Write 18% as a fraction in its lowest terms.

Answer (a) _____ [2]

(b) _____ [2]

3 $\frac{3}{8} \div \frac{1}{16}$;

- (a) write the dividend;
- (b) from the statement above find the quotient.

Answer (a) _____ [1]

(b) _____ [2]

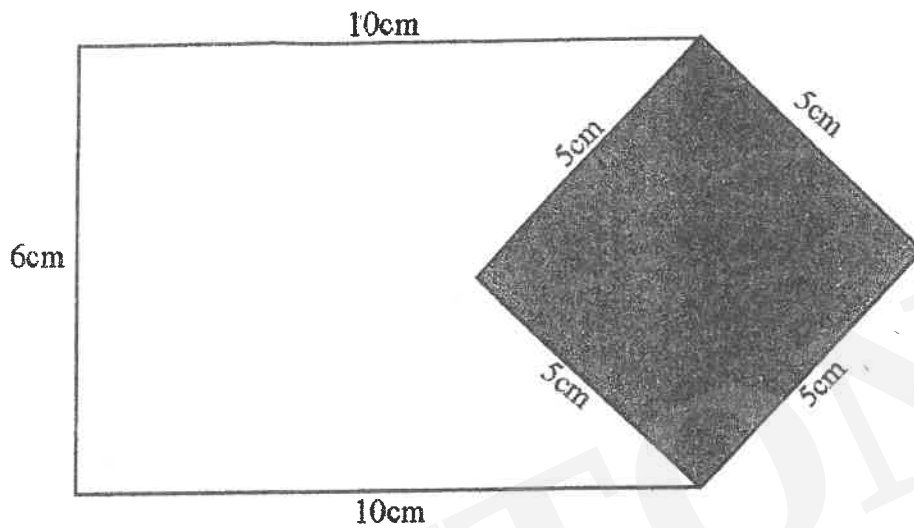
Candidate Name

Centre Number

Candidate Number

- 4 Below is a pentagon with a part shaded.

The shaded part is a square.



- (a) How many lines of symmetry does the shaded part have?
- (b) Find the perimeter of the pentagon.
- (c) Calculate the area of the shaded part.

Answer (a) _____ [1]

(b) _____ [2]

(c) _____ [2]

[Turn over]

Candidate Name

Centre Number

Candidate Number

- 5 Addie earns \$2 700.

The table below shows how Addie spends her salary.

Item	Percentage	Amount
food	30%	\$810
fees	15%	-
rent	15%	-
transport	-	\$810
savings	10%	\$270

- (a) What is the percentage for transport?
- (b) How much savings will Addie have after 5 months.

Answer (a) _____ [2]

(b) _____ [2]

- 6 Simplify $4 + 10 \div 2 - 3 \times 2$

Answer _____ [3]

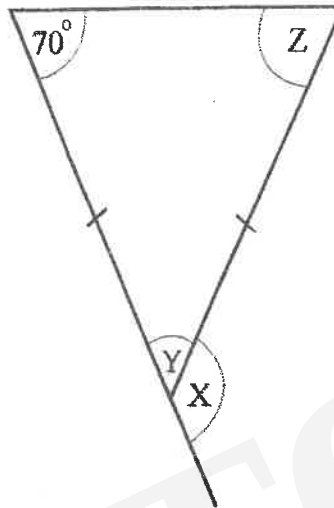
Candidate Name

Centre Number

Candidate Number

7 Study the diagram below.

Answer the questions below.



- (a) Which of the angles in the diagram is obtuse?
- (b) Calculate the angle marked X.

Answer (a) _____ [1]

(b) _____ [3]

[Turn over

Candidate Name

Centre Number

Candidate Number

Section B (15 marks)*Answer three questions in this section.**Each question carries 5 marks.*

- 8 (a) Write 24 October 2013 in Standard International (S.I) notation.
- (b) Mr Brown started a 500 kilometre journey at 6.30 am.
He arrived at 7.00 pm.
- (i) Calculate the time taken to complete the journey.
- (ii) What is his average speed in kilometres per hour?

Answer (a) _____ [1]

(b)(i) _____ [2]

(b)(ii) _____ [2]

Candidate Name

Centre Number

Candidate Number

- 9 Below is a table on Mrs Mussa's wealth.

The value of each item is in decimals.

Item	Value
cattle	0,2
land	0,25
vehicles	0,1
Other items	0,45

- (a) State the item with the least value.
- (b) Express the value of **other items** as a proper fraction in its lowest terms.
- (c) Mrs Mussa's wealth is worth \$200 000.
Calculate the value of cattle in dollars.

Answer (a) _____ [1]

(b) _____ [2]

(c) _____ [2]

[Turn over

Candidate Name

Centre Number

Candidate Number

- 10 Study the fare table below for a Bus Company.

The fares are \$2,50 per kilometre.

Sina					
\$75	Mora				
\$100	\$250	Hore			
\$125	\$200	\$40	Oli		
\$50	\$127,50	\$400	\$600	Came	
\$500	\$200	\$410	\$20	\$750	Cara

- (a) What is the most expensive distance?
- (b) Paul travelled from Came to Sina.
He then went to Cara.
Calculate the total bus fare he paid.
- (c) Find the distance from Hore to Came.

Answer (a) _____ [1]

(b) _____ [2]

(c) _____ [2]

Candidate Name

Centre Number

Candidate Number

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- 11** The Sakala family invested \$6 000 for 42 months.

The simple interest rate was 10% per annum.

- (a) Calculate the simple interest they got.
- (b) Calculate the Sakala family's account balance after 42 months.

Answer (a) _____ [3]

(b) _____ [2]

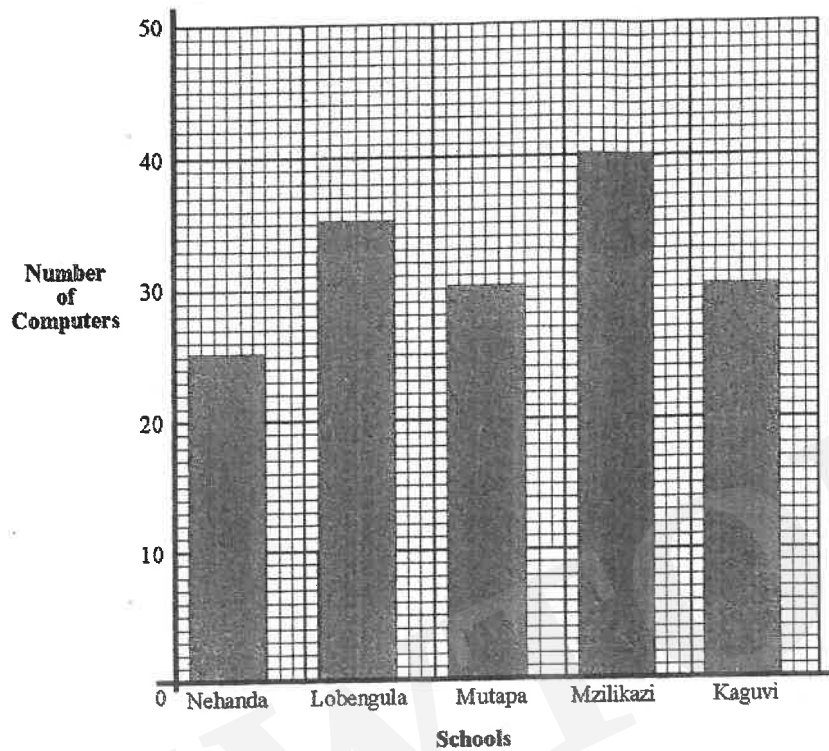
[Turn over

Candidate Name

Centre Number

Candidate Number

- 12 The graph below shows the number of computers in 5 different schools.



- (a) Name the schools with equal number of computers.
- (b) Calculate the total number of computers in the five schools.
- (c) What is the average number of computers for the five schools?

Answer (a) _____ [1]

(b) _____ [2]

(c) _____ [2]

Candidate Name

Centre Number

Candidate Number



For Performance Measurement

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

GRADE SEVEN EXAMINATION, 2023.

MATHEMATICS
PAPER 2

7020/2

OCTOBER 2023 SESSION

Time: 2 hours

Candidates answer on the question paper.

Electronic calculators must not be used.

INSTRUCTIONS TO CANDIDATES

1. Write your name, centre number and candidate number on the spaces at the top of each page.
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7. Do **not** measure from given diagrams.
8. Electronic calculators and slide rules **must not** be used in the examination.

FOR EXAMINER'S USE

Section A

1	
2	
3	
4	
5	
6	
7	
8	
Section B	
TOTAL	

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[Turn over



Candidate Name

Centre Number

Candidate Number



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

GRADE SEVEN EXAMINATION, 2023.

MATHEMATICS
PAPER 2

7020/2

OCTOBER 2023 SESSION

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FOR EXAMINER'S USE

Section A

1	
2	
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Section B	
TOTAL	

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[Turn over



Candidate Name

Centre Number

Candidate Number

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Section A (25 marks)*Answer **all** questions in this section.*

- 1 (a) Round off 40 752 to the nearest thousand.
- (b) What is the value of 9 in 401,98?

Answer: (a) _____ [1]

(b) _____ [1]

- 2 At a school 350 learners are athletes. The ratio of those who play volleyball, soccer and netball is 5: 3: 2, respectively.
- (a) Which sporting discipline has the least share?
- (b) How many soccer players are at the school?

Answer: (a) _____ [1]

(b) _____ [2]



Candidate Name

Centre Number

Candidate Number

--	--	--

- 3 Below is a bank statement for a farmer.

DATE	DEPOSIT	WITHDRAWAL	BALANCE
22 – 09 – 19	\$1 000		\$1 000
28 – 09 – 19		\$850	-
10 – 10 – 19	-		\$650

- (a) Calculate the balance on 28 September.
- (b) How much did the farmer deposit on 10th of October?

Answer: (a) _____ [2]

(b) _____ [2]

- 4 Below is a list of fractions.

$\frac{1}{10}$; $\frac{2}{10}$; $\frac{3}{10}$; $\square$ $\frac{5}{10}$; $\frac{6}{10}$

- (a) Write down the missing fraction.
- (b) Find the sum of the smallest fraction and the biggest fraction in the list.

Answer: (a) _____ [1]

(b) _____ [2]

[Turn over



Candidate Name	Centre Number	Candidate Number

- 5 Below is a table of a full calendar month with week day totals.

Sun	Mon	Tue	Wed	Thur	Fri	Sat
4	5	4	4	4	4	4

- (a) How many days are in this month?
- (b) A girl works from Monday to Friday. She earns \$15 per day.
How much did she earn in the month?

Answer: (a) _____ [2]
(b) _____ [3]

- 6 (a) The factors of 19 are 1 and 19 and factors of 38 are 1, 2, 19 and 38.

State the Highest Common Factor (HCF) of 19 and 38.

- (b) Find the difference between 60,076 and 8,07.

Answer: (a) _____ [1]
(b) _____ [2]



Candidate Name

Centre Number

Candidate Number

--	--	--

- 7 (a) Write the time 0009 in 12 hour notation.
- (b) 8 learners sweep a classroom in 15 minutes.

Find the time taken by 3 learners to sweep the classroom working at the same rate.

Answer: (a) _____ [1]

(b) _____ [2]

- 8 How many grammes make up 2,75 kilogrammes?

Answer: _____ [2]

[Turn over



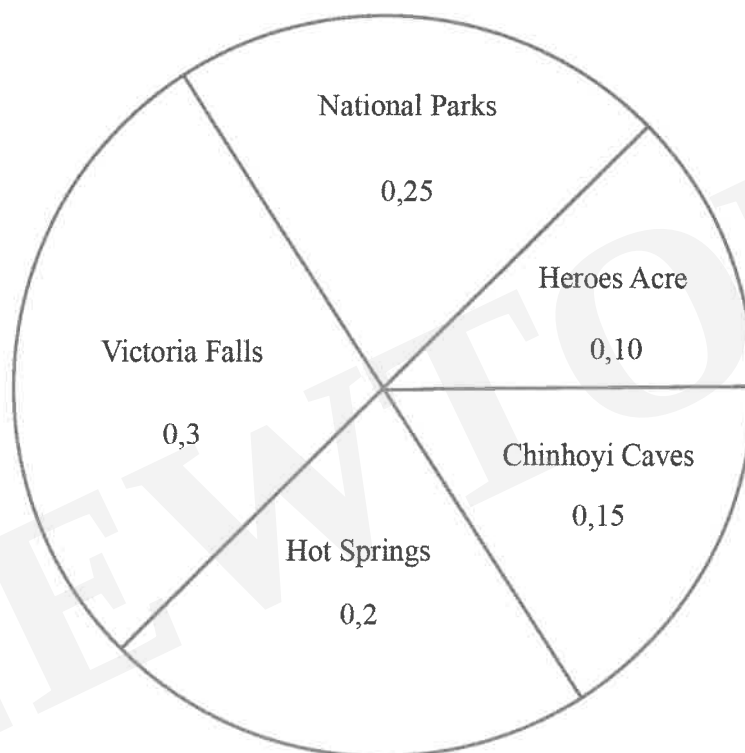
Candidate Name

Centre Number

Candidate Number

Section B (15 marks)*Answer any **three** questions from this section.*

- 9 The pie chart below shows places visited by tourists during the Christmas holiday in 2010, expressed as decimals.



- (a) Write the total number of tourists who visited the Victoria Falls and National Parks as a fraction in its lowest terms.
- (b) Express the number of tourists who visited the National Parks as a percentage.

Answer: (a) _____ [3]

(b) _____ [2]



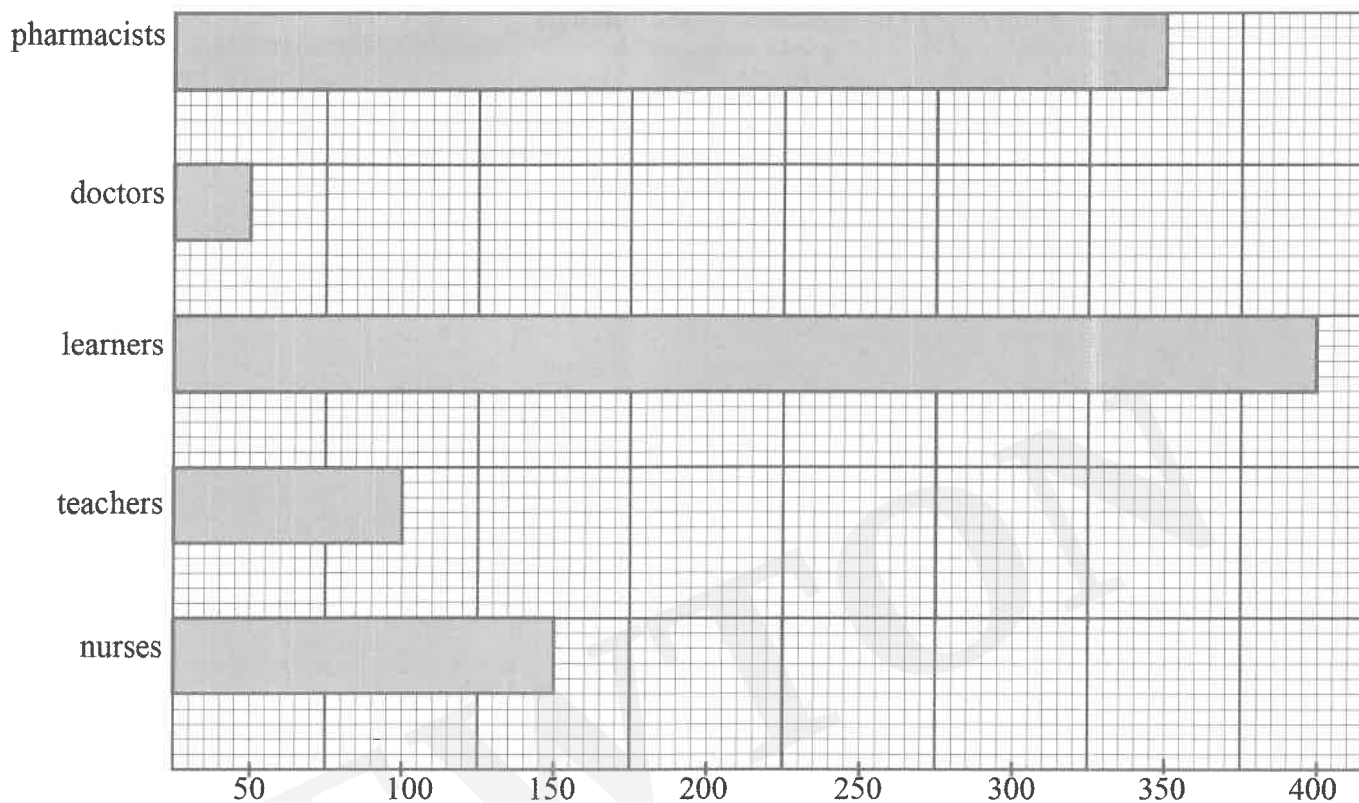
Candidate Name

Centre Number

Candidate Number

--	--	--

- 10 The graph shows the number of people who boarded a local bus in June.



- (a) Find the total number of people who used the bus in June.
- (b) In July $\frac{1}{5}$ of those who used the bus in June used other means of transport.
Find the number of people who used the bus in July.

Answer: (a) _____ [2]

(b) _____ [3]

[Turn over



Candidate Name	Centre Number	Candidate Number

- 11 (a) Write 80 179 in expanded **index** notation.
- (b) A farmer had 640 chickens. $\frac{3}{8}$ were sold.

How many chickens did he remain with?

Answer: (a) _____ [2]

(b) _____ [3]



Candidate Name

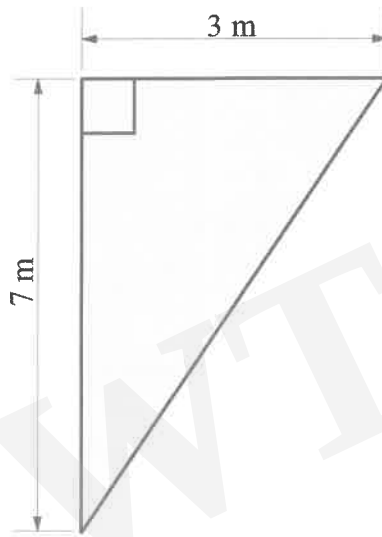
Centre Number

Candidate Number

- 12 (a) A girl measured the length of a fowl run wall and got 18 spans. Her span is 17,5 cm long.

How long is the wall in metres.

- (b) Below is a play centre for a Children's Home.



Calculate the area of the play centre.

Answer: (a) _____ [3]

(b) _____ [2]

[Turn over

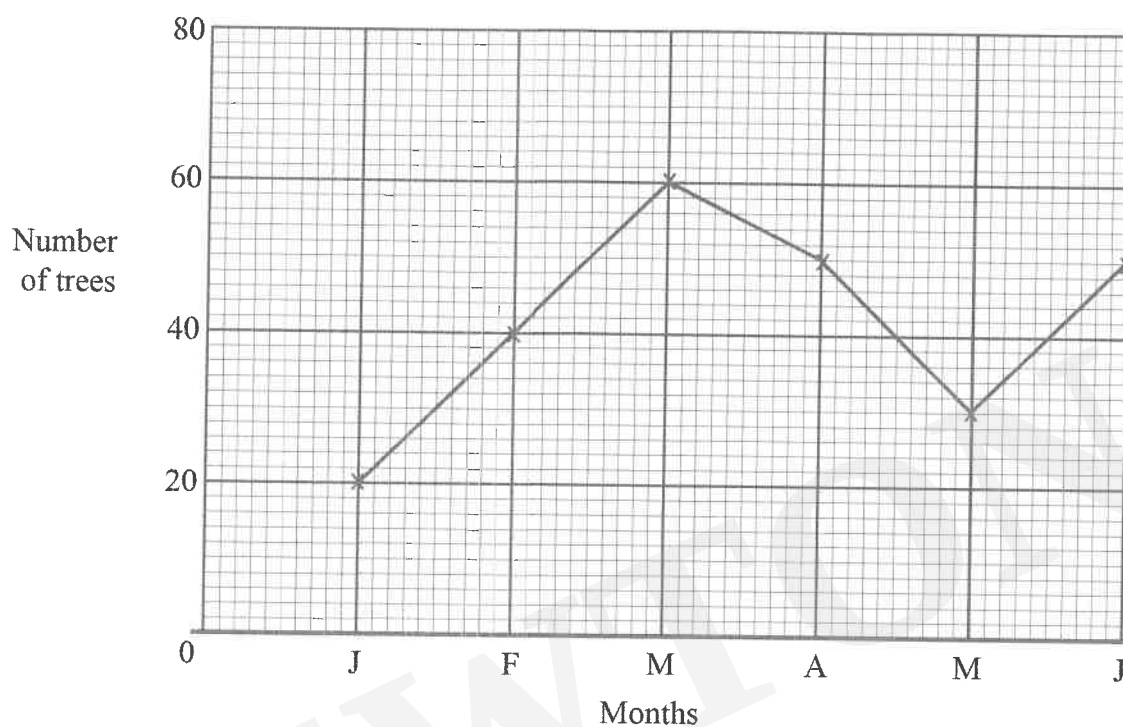


Candidate Name

Centre Number

Candidate Number

- 13 The graph below shows the number of trees planted in a farm from January to June.



- (a) During which month were most trees planted?
- (b) How many more trees were planted from April to June than January to March?

Answer: (a) _____ [1]

(b) _____ [4]





ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
GRADE SEVEN EXAMINATION, 2023.

MATHEMATICS
PAPER 1

7020/01

OCTOBER 2023 SESSION

Time: 2 hours

INSTRUCTIONS TO CANDIDATES

1. Read **all** the instructions carefully.
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6. If **more than one** box is shaded for any one answer, that answer will be regarded as **wrong**.
7. **If you do not understand** the instructions, **ask** the invigilator to explain them to you **before you start**.
8. Answer **all** the questions on the separate answer sheet provided.
9. Paper for rough work will be provided.

This question paper consists of 9 printed pages and 3 blank pages.

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★1. One hundred and ten in numerals is

- A 110.
- B 1010.
- C 1001.
- D 10010.

2. $8 - 4 + 7 =$

- A 3.
- B 5.
- C 11.
- D 19.

★3. The value of the underlined digit in 562,7 is

- A 2.
- B 20.
- C 200.
- D 2 000.

★4. 9 plus 3 equals

- A 3.
- B 6.
- C 12.
- D 27.

★5. 399 rounded off to the nearest ten is

- A 400.
- B 390.
- C 310.
- D 300.

★6. The product of 4 and 12 is

- A 3.
- B 8.
- C 16.
- D 48.

★7. A school assembly started at 0917 and ended at 0935.

How long did the assembly take?

- A 12 minutes
- B 18 minutes
- C 28 minutes
- D 52 minutes

8. $\frac{2}{5}; \frac{1}{2}; \frac{1}{4}; \frac{3}{10}$ arranged in ascending order is

A $\frac{1}{2}; \frac{1}{4}; \frac{2}{5}; \frac{3}{10}$.

B $\frac{1}{4}; \frac{3}{10}; \frac{2}{5}; \frac{1}{2}$.

C $\frac{3}{10}; \frac{2}{5}; \frac{1}{2}; \frac{1}{4}$.

D $\frac{2}{5}; \frac{1}{2}; \frac{3}{10}; \frac{1}{4}$.

9. Which number is exactly divisible by 6 and 9?

A 45

B 3

C 54

D 60

10. $4\frac{3}{7}$ as an improper fraction is

A $\frac{31}{7}$.

B $\frac{28}{7}$.

C $\frac{23}{7}$.

D $\frac{21}{7}$.

11. A triangle with all its sides unequal is called

A right angled.

B equilateral.

C isosceles.

D scalene.

12. 

The tally shown above represents

A 9.

B 19.

C 29.

D 39.

[Turn over



13. Which of the following is a prime number?

- A 20
- B 21
- C 22
- D 23

*14. The area of a square with sides 7 cm is

- A 14 cm^2 .
- B 22 cm^2 .
- C 28 cm^2 .
- D 49 cm^2 .

*15. The Lowest Common Multiple (LCM) of 4 and 8 is

- A 4.
- B 8.
- C 16.
- D 32.

16. A fruit vendor bought 2 boxes of apples at \$300.

If each box had 75 apples, the cost of one apple was

- A \$4.
- B \$3.
- C \$2.
- D \$1.

*17. $0,09 + 0,6 + 8 =$

- A 9,68.
- B 8,69.
- C 8,15.
- D 1,49.



Use the table below to answer questions 18 and 19.

The table shows the number of patients that visited a clinic in five months.

Month	Number of patients
May	160
June	192
July	201
August	325
September	142

18. How many more patients visited the clinic in July than June?
- A 9
B 19
C 119
D 191
19. What is the total number of people who visited the clinic from May to September?

- A 878
B 920
C 1 019
D 1 020

20. Noma's bank account had \$75,97. She deposited \$239,50.

What was her new balance?

- A 163,53
B 164,43
C 314,47
D 315,47

21. $\frac{1}{4} \div \frac{1}{3} =$

- A $\frac{1}{12}$
B $\frac{2}{7}$
C $\frac{7}{12}$
D $\frac{3}{4}$

[Turn over



*22. The value of six \$5 notes, three \$2 notes, two 50 cent coins and three 25 cent coins is

- A \$36,75.
- B \$37,75.
- C \$110,00.
- D \$112,00.

*23. $\frac{7}{8}$ expressed as a percentage is

- A 87,5%.
- B 85,5%.
- C 19,5%.
- D 12,5%.

*24. How many degrees are in $\frac{3}{8}$ of a revolution?

- A 45
- B 125
- C 135
- D 225

*25. 3,254 in expanded notation is

- A 3 000 + 200 50 + 4.
- B 3 200 + 50 + 4.
- C $3 + \frac{2}{10} + \frac{5}{100} + \frac{4}{1000}$.
- D $4 + \frac{5}{10} + \frac{2}{100} + \frac{3}{1000}$.

*26. An empty ox-drawn wagon weighs 174 kg.

If it carries ten 50 kg bags of maize, the gross mass in kilogrammes will be

- A 326.
- B 500.
- C 436.
- D 674.



*27. $\frac{16}{6}$ expressed as a mixed number is

- A $2\frac{5}{6}$.
- B $2\frac{2}{3}$.
- C $2\frac{1}{2}$.
- D $2\frac{1}{3}$.

*28. The volume of a rectangular prism measuring 8 cm by 5 cm by 4 cm is

- A 160 cm^3 .
- B 144 cm^3 .
- C 40 cm^3 .
- D 17 cm^3 .

*29. The cost of one loaf of bread is \$6,50.

What is the cost of one dozen of bread?

- A \$65,00
- B \$65,10
- C \$78,00
- D \$78,12

30. At a camp 36 and 48 people slept in tents and hall respectively.

What was the ratio of people who slept in a hall as to those in tents in its lowest terms?

- A 8 : 6
- B 6 : 8
- C 4 : 3
- D 3 : 4

31. What is the difference between $6 \times (9 + 8)$ and $6 \times 9 + 8$?

- A 164
- B 62
- C 40
- D 0

[Turn over



Use the table below to answer questions 32 and 33.

The table shows the time taken by 4 students to finish a race.

Student	Nothando	Rudo	Thulani	Ravai
Time in minutes	5,5	3,4	4,6	8,2

32. Who came first?

- A Nothando
- B Thulani
- C Ravai
- D Rudo

33. The difference between Ravai and Rudo's times is

- A 4,8.
- B 5,2.
- C 11,6.
- D 12,6.

34. A family owns one hectare of land. They planted cabbages on $\frac{2}{5}$ of the land.

How many square metres are yet to be planted?

[1 hectare = 10 000 square metres]

- A 1 000
- B 2 000
- C 4 000
- D 6 000

Use the calendar below to answer questions 35 and 36.

M		6	13	20	27
T		7	14	21	28
W	1	8	15	22	29
TH	2	9	16	23	30
F	3	10	17	24	
S	4	11	18	25	
S	5	12	19	26	

*35. On which day was the 25th of the month?

- A Thursday
- B Friday
- C Saturday
- D Sunday



★36. How many Wednesdays were there in the month of April?

- A 4
- B 5
- C 6
- D 7

37. Mr Mheni bought five computers at \$200 each. He was given 10% discount.

The amount paid was

- A \$1 100.
- B \$1 000.
- C \$900.
- D \$100.

★38. How many years are there in a decade?

- A 10
- B 50
- C 100
- D 1 000

39. The ratio of girls to boys in a school is 5: 3. The total number of children in the school is 960.

What is the number of girls in the school?

- A 620
- B 600
- C 125
- D 120

40. Lona is facing South West. She turns three right angles anticlockwise.

Now she faces

- A North-West.
- B South-West.
- C South-East.
- D North-East.



Candidate Name

Centre Number

Candidate Number



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

GRADE SEVEN EXAMINATION, 2023.

MATHEMATICS PAPER 2

7020/2

OCTOBER 2023 SESSION

Time: 2 hours

Candidates answer on the question paper.

Electronic calculators must not be used.

INSTRUCTIONS TO CANDIDATES

1. Write your name, centre number and candidate number on the spaces at the top of each page.
2. Answer **all** questions in Section A and any **three** questions in Section B.
3. If more than **three** questions are answered in **Section B**, the first three will be considered.
4. If working is needed for any question it must be shown in the space below that question.
5. Answers must be written in the spaces indicated below each question.
6. Omission of essential working may result in loss of marks.
7. Do **not** measure from given diagrams.
8. Electronic calculators and slide rules **must not** be used in the examination.

FOR EXAMINER'S USE

Section A

1	
2	
3	
4	
5	
6	
7	
8	

Section B

TOTAL	

This question paper consists of 10 printed pages and 2 blank pages.

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[Turn over]



Candidate Name

Centre Number

Candidate Number

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Section A (25 marks)*Answer **all** questions in this section.*

1 (a) Round off 40 752 to the nearest thousand.

(b) What is the value of 9 in 401,98?

Answer: (a) _____ [1]

(b) _____ [1]

2 At a school 350 learners are athletes. The ratio of those who play volleyball, soccer and netball is 5: 3: 2, respectively.

(a) Which sporting discipline has the least share?

(b) How many soccer players are at the school?

Answer: (a) _____ [1]

(b) _____ [2]



Candidate Name

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- 3 Below is a bank statement for a farmer.

DATE	DEPOSIT	WITHDRAWAL	BALANCE
22 – 09 – 19	\$1 000		\$1 000
28 – 09 – 19		\$850	-
10 – 10 – 19	-		\$650

- (a) Calculate the balance on 28 September.
- (b) How much did the farmer deposit on 10th of October?

Answer: (a) _____ [2]

(b) _____ [2]

- 4 Below is a list of fractions.

$\frac{1}{10}$; $\frac{2}{10}$; $\frac{3}{10}$; $\square$ $\frac{5}{10}$; $\frac{6}{10}$

- (a) Write down the missing fraction.
- (b) Find the sum of the smallest fraction and the biggest fraction in the list.

Answer: (a) _____ [1]

(b) _____ [2]

[Turn over



Candidate Name

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- 5 Below is a table of a full calendar month with week day totals.

Sun	Mon	Tue	Wed	Thur	Fri	Sat
4	5	4	4	4	4	4

- (a) How many days are in this month?
- (b) A girl works from Monday to Friday. She earns \$15 per day.
How much did she earn in the month?

Answer: (a) _____ [2]

(b) _____ [3]

- 6 (a) The factors of 19 are 1 and 19 and factors of 38 are 1, 2, 19 and 38.

State the Highest Common Factor (HCF) of 19 and 38.

- (b) Find the difference between 60,076 and 8,07.

Answer: (a) _____ [1]

(b) _____ [2]



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- 7 (a) Write the time 0009 in 12 hour notation.
- (b) 8 learners sweep a classroom in 15 minutes.

Find the time taken by 3 learners to sweep the classroom working at the same rate.

Answer: (a) _____ [1]

(b) _____ [2]

- 8 How many grammes make up 2,75 kilogrammes?

Answer: _____ [2]

[Turn over



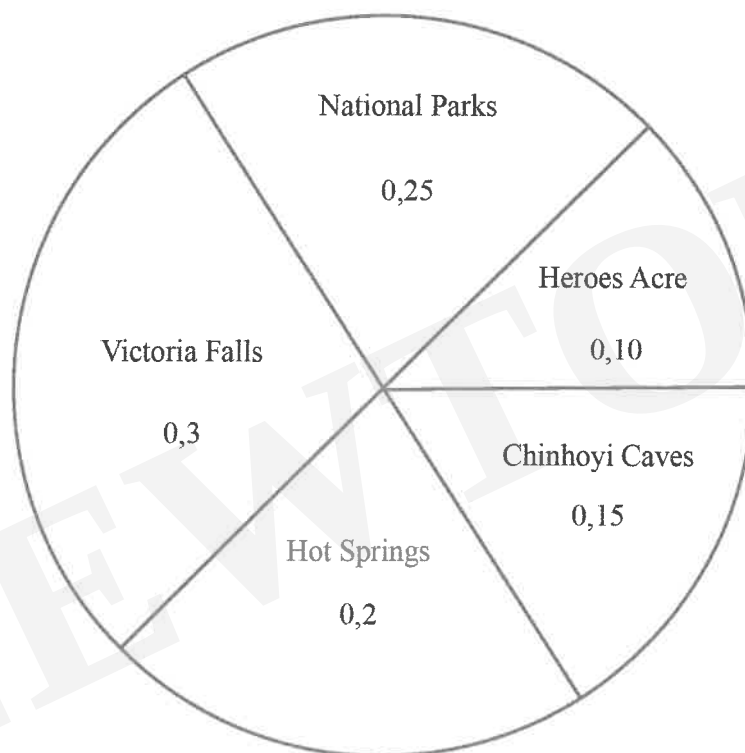
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Candidate Number

Section B (15 marks)*Answer any **three** questions from this section.*

- 9 The pie chart below shows places visited by tourists during the Christmas holiday in 2010, expressed as decimals.



- (a) Write the total number of tourists who visited the Victoria Falls and National Parks as a fraction in its lowest terms.
- (b) Express the number of tourists who visited the National Parks as a percentage.

Answer: (a) _____ [3]

(b) _____ [2]

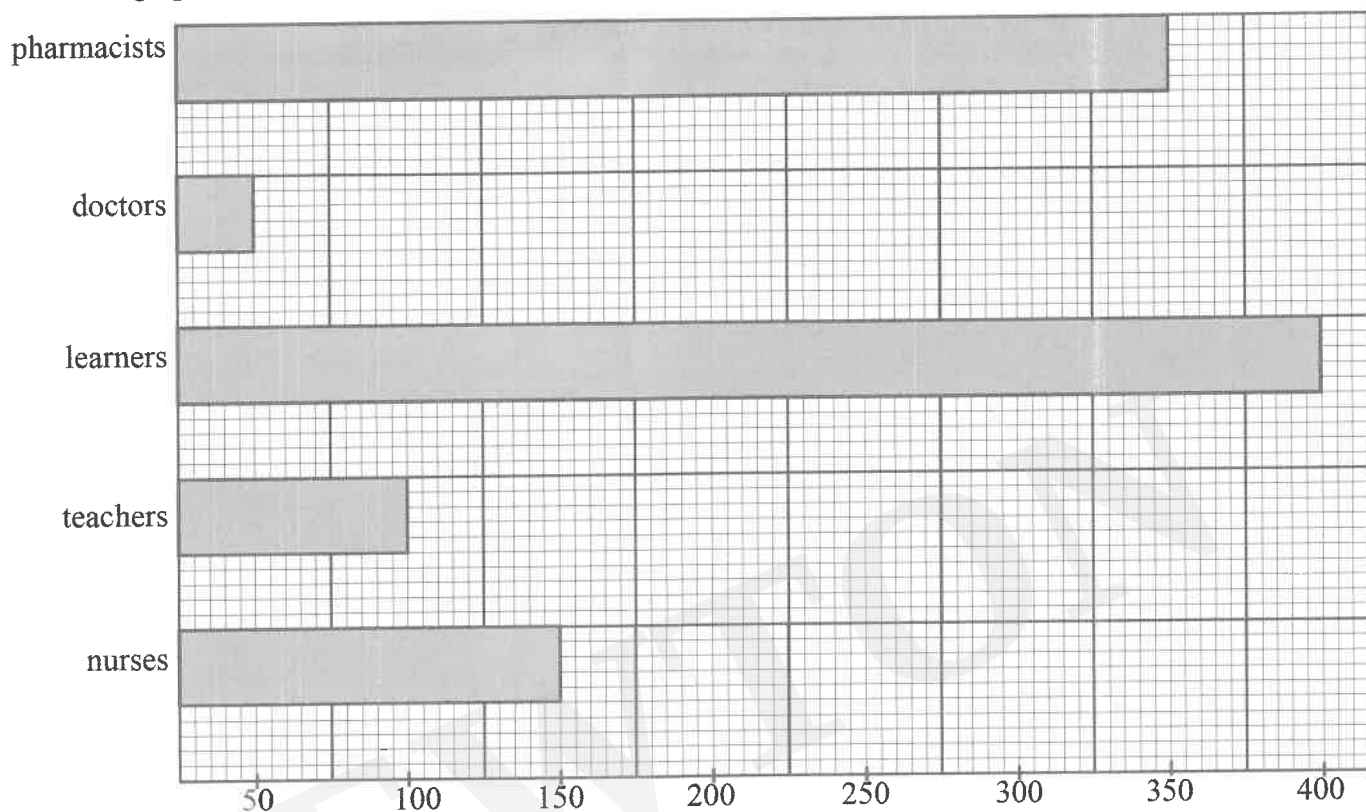


Candidate Name

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- 10 The graph shows the number of people who boarded a local bus in June.



- (a) Find the total number of people who used the bus in June.
- (b) In July $\frac{1}{5}$ of those who used the bus in June used other means of transport.
Find the number of people who used the bus in July.

Answer: (a) _____ [2]

(b) _____ [3]

[Turn over



Candidate Name

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11 (a) Write 80 179 in expanded **index** notation.

(b) A farmer had 640 chickens. $\frac{3}{8}$ were sold.

How many chickens did he remain with?

Answer: (a) _____ [2]

(b) _____ [3]



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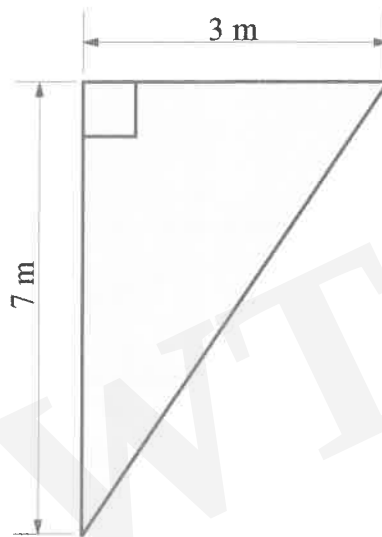
Candidate Number

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- 12 (a) A girl measured the length of a fowl run wall and got 18 spans. Her span is 17,5 cm long.

How long is the wall in metres.

- (b) Below is a play centre for a Children's Home.



Calculate the area of the play centre.

Answer: (a) _____ [3]

(b) _____ [2]

[Turn over

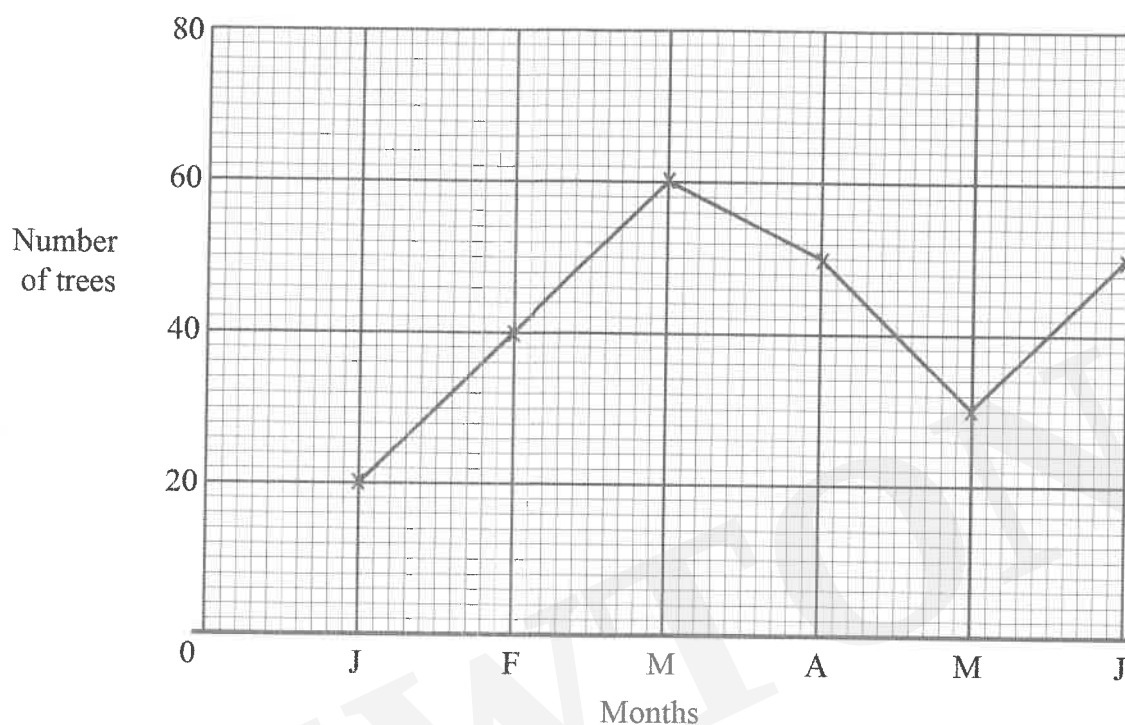


Candidate Name

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- 13 The graph below shows the number of trees planted in a farm from January to June.



- (a) During which month were most trees planted?
- (b) How many more trees were planted from April to June than January to March?

Answer: (a) _____ [1]

(b) _____ [4]

